Measuring, Localizing, and Ablating Alignment Signatures in LLMs

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Abstract

Aligned language models often exhibit a recognizable AI-like style, yet its connection to post-training and internal representations remains poorly understood. In this work, we study whether post-training introduces or amplifies AI-like stylistic regularities and whether these regularities have a localized internal signature. To this end, we compare human text, base-model generations, and aligned-model generations under matched human-source prefixes. Aligned generations show lower human-corpus affinity and higher AI-detection rates than base generations, suggesting that post-training shifts generated text away from human-corpus style and toward detector-visible AI-like text. We then introduce **PASTA** (**P**ost-training **A**lignment **S**ignature **T**argeted **A**blation), a training-free method that estimates a post-training alignment signature from aligned-base residual contrasts and ablates the corresponding direction during decoding. Across 11 aligned models and 6 AI-detectors, PASTA lowers the detection rate for most aligned models; this effect transfers well across detectors and is not reproduced by random directions. Qualitative analysis suggests that PASTA generations remain relevant and coherent while exhibiting greater stylistic variation. Together, these results show that AI-like stylistic effects of post-training can be measured, localized, and causally tested through activation ablation.[1]

1 Introduction

Aligned language models produce text with a recognizable style: polished phrasing, hedging, affirming phrases, and highly regular structure (Chakrabarty et al., 2025; Russell et al., 2026). While these patterns are familiar at the output level, they raise a deeper question: *To what extent are AI-like stylistic regularities introduced by post-training, and how are they represented inside the model?* Prior work has shown that post-training changes output diversity, expressed opinions, verbosity, and behaviors such as sycophancy (Kirk et al., 2024; Santurkar et al., 2023; Perez et al., 2023; Sharma et al., 2024). However, these studies characterize post-training effects behaviorally, focusing on output-level changes. It remains unclear whether the recognizable AI-like style itself is a post-training effect and whether a meaningful component of this effect can be localized in, and ablated from, aligned-model representations.

To address this gap, we contrast aligned models with their base counterparts under matched prompting conditions. This contrast isolates stylistic regularities introduced or amplified by post-training from those already present after pretraining. We then ask whether this post-training effect is only an output-level distributional shift, or whether it corresponds to an identifiable structure inside the model. If the effect is diffuse across representations, it may be difficult to localize or intervene on. If, instead, a meaningful component is captured by a single residual-stream direction, then post-training leaves a concrete representational footprint. We call this direction a *post-training alignment signature*. Building on work that links behaviors such as refusal to linear directions in representation space (Arditi et al., 2024; Zou et al., 2023), we ask whether broader AI-like stylistic regularities induced by post-training can be localized and causally tested in the same way.

This leads to two empirical questions: 1) Does post-training induce an output-level shift from base to aligned generations? 2) If a shift exists, can it be localized to an identifiable direction in the residual stream? To answer the first question, we use matched prefixes to compare human text, base-model generations, and aligned-model generations. Aligned generations show lower affinity to human-corpus indexes under infini-gram (Liu et al., 2024)

[1]**Code+Data:** github.com/alignment-signature/alignment-signature.

and higher AI-detection rates than base generations, suggesting that post-training makes generated text less human-corpus-like and more detector-visible. We use detectors as probes of stylistic separability rather than training objectives.

To answer the second question, we introduce **PASTA** (**P**ost-training **A**lignment **S**ignature **T**argeted **A**blation), a training-free method that estimates an aligned-minus-base residual direction and removes it during aligned-model decoding. A detector-guided model layer sweep selects the direction used for ablation. Across 11 aligned models and six detectors, PASTA lowers detector flagging for most models, with Pangram (Emi and Spero, 2024) reductions exceeding 90 percentage points for several models. The effect transfers to detectors not used for layer selection and is not reproduced by random directions, suggesting that the extracted direction captures a specific post-training signal. Qualitative analysis suggests that PASTA generations remain relevant and coherent while exhibiting greater stylistic variation.

Our contributions are twofold. First, we provide output-level evidence that post-training shifts aligned generations away from human-corpus style and toward detector-visible AI-like text. Second, we provide evidence that post-training leaves an identifiable representation-level signature associated with AI-like stylistic regularities in generated text. We introduce PASTA, a training-free intervention that extracts this signature from aligned-base activation contrasts and projects it out during aligned-model decoding. Together, these results show that AI-like stylistic effects of post-training can be measured, localized, and causally tested through interventions on model activations.

2 Related Work

Activation-Space Interventions and Post-Training Shifts. One line of work shows that behaviors can be localized and edited in activation space. Representation Engineering (Zou et al., 2023), Activation Engineering (Turner et al., 2023), Inference-Time Intervention (Li et al., 2023), and Contrastive Activation Addition (Rimsky et al., 2024) steer model behavior at inference time by adding or removing components along activation directions, without updating weights. These methods have been used to steer behaviors such as truthfulness, sentiment, and compliance. Most closely related, Arditi et al. (2024) identify a single residual-stream direction mediating refusal behavior in aligned models. A second line of work shows that post-training reshapes model outputs beyond task performance. Instruction tuning and RLHF (Ouyang et al., 2022; Bai et al., 2022) affect expressed opinions and political leaning (Santurkar et al., 2023), sycophancy and reward-driven behaviors (Perez et al., 2023; Sharma et al., 2024; Wei et al., 2023), verbosity and length bias (Saito et al., 2023; Singhal et al., 2024), output diversity (Kirk et al., 2024), and recognizable writing-style artifacts (Chakrabarty et al., 2025; Russell et al., 2026). These studies primarily characterize post-training effects behaviorally. Our work connects these two lines of work: we ask whether an AI-like stylistic effect of post-training is not only visible in outputs but also localized in model activations and causally testable through ablation.

Base–Aligned Contrasts and AI-Text Detectors. Prior work compares base and aligned variants to isolate what post-training adds. Lin et al. (2024) show that prompting can recover much of aligned models' surface behavior from base models; Fei et al. (2025) switch between base and aligned decoding using base-model uncertainty; Gui et al. (2024) analyze best-of-n alignment as a proxy for post-training effects; and Wang et al. (2025) mix base and aligned generations to recover output diversity. We also use the base model as a reference point, but focus on localizing the post-training shift rather than mixing or switching between models. We use AI-text detectors as measurement probes for this shift. Existing detectors include likelihood-based methods (Mitchell et al., 2023; Bao et al., 2024), observer-model approaches such as Binoculars (Hans et al., 2024), rewriting-based methods such as RAIDAR (Mao et al., 2024), and watermarking (Kirchenbauer et al., 2023). Since detector behavior varies across models, decoding settings, and domains (Tufts et al., 2025), we treat detector outputs as signals of stylistic separability, not ground-truth authorship labels. Prior work has also shown that paraphrasing, rewriting, and stylometric obfuscation can reduce detector confidence (Krishna et al., 2023; Tripto et al., 2024; Xing et al., 2024; Lu et al., 2024; Cheng et al., 2025; Fang et al., 2025). In contrast, we use reduced detector visibility to study a post-training representation-level signature, not to claim robust detector evasion.

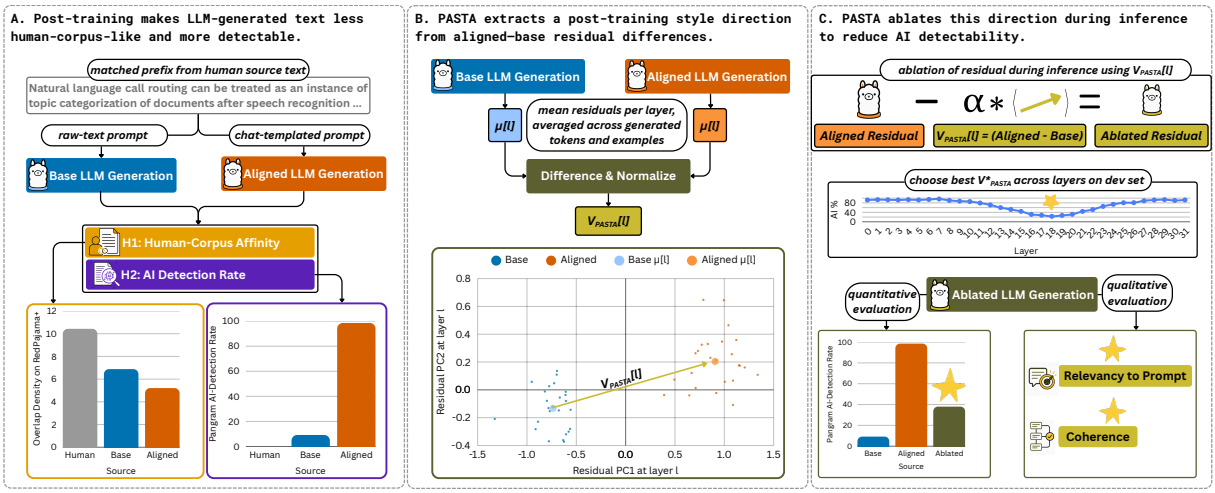


Figure 1: **PASTA overview. (A)** Under matched human-sourced prefixes, aligned-model generations show lower infini-gram affinity to RedPajama+ and higher AI-detection rates than base-model generations, suggesting that post-training shifts generated text away from human-corpus style and toward detector-visible text. **(B)** PASTA estimates this shift as a layer-wise residual-stream direction by averaging residual states over generated tokens and examples, then taking the normalized aligned-minus-base difference. **(C)** A detector-guided layer sweep selects the most effective ablation direction, which PASTA projects out during inference. The resulting ablated generations have lower AI detectability while maintaining comparable prompt relevance and coherence in qualitative evaluations.

3 Post-Training Shifts Generations Away from Human-Corpus Text

Post-training optimizes base models toward a much narrower text distribution than web-scale pretraining: instruction and preference data are curated for helpful, polished, structured, and rater-preferred responses. We ask whether this narrower optimization target leaves a surface-level signature in generation, beyond improving instruction following. Specifically, *does post-training move outputs away from naturally occurring human text and toward a more recognizable assistant-like style?*

Setup. We test the post-training shift with matched triples:

$$(t_{\text{human}},\ t_{\text{base}},\ t_{\text{aligned}}),$$

constructed for each source domain and base/aligned model pair. The domains are scientific abstracts, creative fiction, opinion pieces, college essays, and news articles. Each model pair consists of a pretrained base model and its post-trained counterpart, such as Llama-3.1-8B and Llama-3.1-8B-Instruct. For each triple, t_{human} is the original human continuation, while t_{base} and t_{aligned} are generated from the same human-written prefix under matched prompting. To avoid length as a confound, we length-control all continuations before evaluation: base and aligned generations are retained only if their character length is within $\pm 20\%$ of the corresponding human continuation. By fixing the prefix, domain, model family, and length, the base–aligned contrast isolates the effect of post-training. Appendix A details the domain sources and prefix construction; Appendix B details the models and generation settings; Appendix B.2 details prompt formats.

Approach. We probe the post-training surface shift with two complementary hypotheses. **H1** tests whether aligned outputs have lower verbatim affinity to a pretraining-style human reference corpus than base outputs. **H2** tests whether aligned outputs receive higher AI-detector scores than base outputs. These probes are not complete measures of human-likeness; they capture complementary properties. Corpus affinity measures proximity to naturally occurring human text, while detector scores capture detector-visible AI-generated regularities.

H1: Post-training reduces verbatim source affinity. Our first hypothesis tests whether post-training moves model generations away from naturally occurring human text. We measure this using verbatim affinity to a large reference corpus. For a reference corpus C and a text $t = (w_1, \ldots, w_N)$, let ℓ_i be the length, in words, of the longest contiguous span beginning at position i that appears at least once in C. We define the *overlap density* of t

as

$$\rho_C(t) = \frac{1}{N} \sum_{i=1}^{N} \ell_i, \tag{1}$$

where higher values indicate longer corpus-matched spans across the text. We also report Coverage@10, the fraction of words covered by corpus-matched spans of at least 10 words, as a long-span robustness check.

Let ρ_{H}, ρ_{B}, and ρ_{A} denote the mean overlap density for human, base, and aligned continuations, respectively. H1 predicts

$$\rho_{\mathrm{H}} > \rho_{\mathrm{B}} > \rho_{\mathrm{A}}. \tag{2}$$

The first inequality reflects that human continuations are drawn from the source distribution. The second is the key post-training test: if post-training pushes models toward a narrower assistant-style distribution, aligned generations should have lower affinity to pretraining-style human text than base generations from the same prefixes.

We instantiate C with a RedPajama+ infini-gram index (details in Appendix C) over pre-2023 publicly available text. The index is a proxy for broad pretraining-style human text, not evidence of exact membership in any model's training data. Figure 2 shows normalized corpus-affinity scores across scientific abstracts, creative fiction, opinion pieces, college essays, and news articles, with each domain divided by its human-continuation mean. Both overlap density and Coverage@10 follow Human $>$ Base $>$ Aligned in every domain. The drop is largest when the index has close source overlap with the domain; opinion pieces show a weaker but directionally consistent effect. Thus, aligned generations have lower verbatim affinity to pretraining-style human text than base generations from the same human-written prefixes, supporting H1. Full raw scores, normalized scores and metric definitions are reported in Appendix C.1.

H2: Post-training increases AI-detector visibility. H2 asks whether the surface shift in H1 is also visible to AI detectors. Let $d(t)$ denote an AI-detector score or classification rate, where larger values mean that t is more likely to be classified as AI-generated. Let d_{H}, d_{B}, and d_{A} denote the mean detector score for human, base, and aligned continuations, respectively. H2 predicts

$$d_{\mathrm{H}} < d_{\mathrm{B}} < d_{\mathrm{A}}. \tag{3}$$

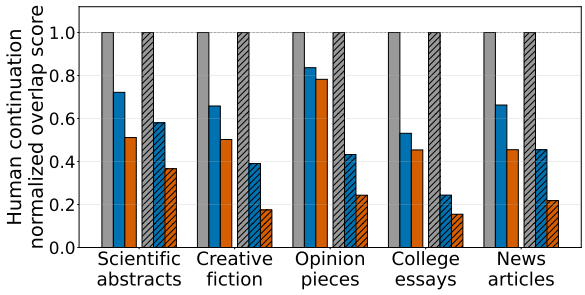


Figure 2: **Post-training reduces verbatim affinity to human-corpus text.** For each domain, we compare matched continuations generated from the same human-written prefix: Human, Base, and Aligned. Scores are normalized by the human-continuation mean within each domain, so Human $= 1.0$ by construction. Solid bars show overlap density, the mean longest RedPajama+-matched span across word positions; hatched bars show Coverage@10, the fraction of words covered by RedPajama+ matched spans of at least 10 words. Across domains and metrics, scores follow Human $>$ Base $>$ Aligned, indicating that post-training moves generations further away from source-domain human text.

The key comparison is t_{base} versus t_{aligned}, since both are generated from the same prefix by models from the same family. If aligned continuations receive higher detector scores than base continuations, the increase is not merely a property of machine-generated text. It indicates that post-training adds detector-visible regularities on top of the base-model distribution.

We evaluate H2 on same models and domains as H1. To make the test conservative, we filter base-model continuations for degeneracy and re-rank them with a Claude-Sonnet-4.6 judge for detecting degeneracy. Thus, each aligned output is compared against a base output selected to be non-degenerate.

Figure 3 reports AI-detection rates for base and aligned generations across detectors, ordered by accuracy on our detector-quality evaluation (Appendix D.2). Nearly all detectors flag aligned generations more often than base generations. The largest gaps are from Pangram and GPTZero: Pangram flags 98.5% of aligned outputs versus 9.2% of base outputs, while GPTZero flags 84.5% versus 2.9%. Fast-DetectGPT is the only exception; which is unreliable, indicated by a higher false-positive rate on human text. These results support H2: under matched prefixes, lengths, and model families, aligned generations are more detector-visible than base generations. The effect is strongest for higher-accuracy detectors, suggesting that post-training

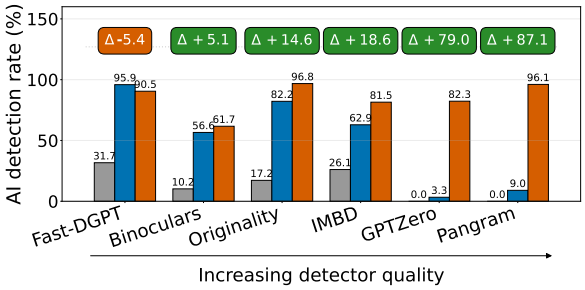


Figure 3: **Post-training increases AI-detector visibility.** We report AI-detection rates for Human, Base, and Aligned continuations. Base and Aligned outputs are generated from the same human-written prefixes and length-controlled before evaluation. Detectors are ordered from left to right by increasing macro accuracy on our detector-quality evaluation. Higher-quality detectors (Appendix D.2) assign much higher AI rates to aligned outputs than to base outputs, supporting H2 that post-training introduces detector-visible regularities.

adds a shared detector-visible style beyond the base-model distribution.

Takeaway: Post-training Leaves a Surface Signature. H1 and H2 show the same shift from complementary angles. Under matched human-written prefixes, aligned generations are less similar to pretraining-style human text and more visible to AI detectors than base generations. This supports Figure 1A: post-training moves generations away from human-corpus-like text and toward detector-visible assistant-style text. This surface signature could reflect different token-level statistics, but it does not identify a responsible mechanism. We next test whether part of it is mediated by a locatable residual-stream direction.

4 PASTA: Post-training Alignment Signature Targeted Ablation

The surface signature in Section 3 shows that post-training moves generations toward a more detector-visible assistant-style regime. We now ask whether part of this shift is mediated by a locatable internal feature. Our hypothesis is that post-training induces a residual-stream direction that separates aligned-model behavior from base-model behavior. If this direction is causally involved in detector-visible style, then ablating it during decoding should reduce AI-detector visibility. We call this intervention PASTA: **P**ost-training **A**lignment **S**ignature **T**argeted **A**blation.

Method. Let $\mathcal{M}_{\text{base}}$ be a pretrained model and $\mathcal{M}_{\text{align}}$ its post-trained counterpart. Given a small calibration set of matched naturalistic prompts, we generate continuations from both models and record residual-stream activations over generated tokens. For each layer ℓ, we compute the mean residual state for each model:

$$\mu^{\ell}_{\text{base}} = \mathbb{E}_{p,t}\left[h^{\ell}_{\text{base}}(p,t)\right], \tag{4}$$

$$\mu^{\ell}_{\text{align}} = \mathbb{E}_{p,t}\left[h^{\ell}_{\text{align}}(p,t)\right], \tag{5}$$

where $h^{\ell}(p,t)$ is the residual-stream activation at layer ℓ for generated token t under prompt p. Their normalized difference defines the candidate post-training direction:

$$v^{\ell}_{\text{PASTA}} = \frac{\mu^{\ell}_{\text{align}} - \mu^{\ell}_{\text{base}}}{\left\|\mu^{\ell}_{\text{align}} - \mu^{\ell}_{\text{base}}\right\|_2}. \tag{6}$$

To test whether this direction is causally involved in detector-visible style, we ablate it during aligned-model decoding. For a residual vector x and a unit direction v, PASTA removes the component of x along v with strength α:

$$\text{Ablate}_{\alpha}(x; v) = x - \alpha(x^{\top}v)v \tag{7}$$

This projection, with default $\alpha = 1$, is applied during autoregressive generation at residual-stream write sites of the aligned model. To choose the most effective v, we first perform a detector-guided layer sweep on the calibration set. For each candidate layer ℓ, we generate ablated continuations using v^{ℓ}_{PASTA} and score them with a detector $\mathcal{D}$. We select the layer ℓ^* whose direction yields the lowest calibration-set AI-detection rate, then fix $v^{\ell^*}_{\text{PASTA}}$ for held-out test prompts. Figure 1 summarizes the pipeline, and Appendix E.1 gives full pseudocode.

Experimental Setup. For each aligned model, we estimate a single PASTA direction using 25 randomly selected examples[2], with five examples from each domain. The direction is selected with Pangram on the pooled calibration set and then applied to 500 held-out examples, with 100 examples per domain. Calibration and test prompts are disjoint. Unless otherwise specified, we use Pangram for detector-guided layer selection because it is the strongest detector in our detector-quality

[2]Using 100 calibration examples yielded directions similar to 25-example directions with a cosine similarity of 0.92–0.96 across 4 models.

evaluation. We evaluate the resulting generations with Pangram, Binoculars, ImBD, and RAIDAR. Pangram measures performance on the selection detector, while the remaining detectors test cross-detector transfer. For each model m and detector d, we report the signed change in AI-detection rate after PASTA ablation:

$$\Delta_{m,d}^{PASTA} = \mathrm{AI}_{m,d}^{PASTA} - \mathrm{AI}_{m,d}^{\text{aligned}},$$

where $\mathrm{AI}_{m,d}^{c}$ is the percentage of generations from condition c flagged as AI by detector d for model m. Negative values indicate reduced detector visibility after PASTA ablation.

Results. Table 1 shows that PASTA reduces detector visibility across many aligned models. The largest reductions occur on Pangram, the detector used for layer selection: 8 of 11 models drop by 73.2–98.0 points. These include Gemma-2-9B, several Qwen2.5 models from 1.5B to 32B, and Llama-3.1-8B-Instruct. The effect also transfers beyond the selection detector. ImBD and RAIDAR decrease for every model, and Binoculars decreases for 10 of 11 models. This cross-detector transfer suggests that PASTA is not merely overfitting to Pangram, but removing a broader detector-visible component of the post-training signature. The weakest cases are OLMo-3-7B-Instruct and Tulu-3-8B-RLVR. These models show the smallest Pangram reductions ($\leq$ 6.0 points), and Binoculars sometimes increases after ablation. We interpret these exceptions cautiously because Appendix D.2 shows that Binoculars is less reliable than the strongest detectors in our setting. More broadly, these failures suggest that the strength and linearity of the post-training signature vary across model families and post-training recipes. Overall, these results support the mechanistic hypothesis behind PASTA. For many aligned models, an aligned-minus-base residual-stream direction captures a detector-visible component of post-training. Ablating this direction at inference time reduces AI-detector visibility, often transfers across detectors, and requires only a small calibration set with no parameter updates.

5 Analyzing PASTA: Specificity, Quality, and Robustness

We ask why PASTA works and when its effects hold. We test whether reduced AI-detectability is specific to the PASTA direction rather than a generic effect of residual-stream ablation. We then assess generation quality and robustness across ablation strengths, domains, and detectors. Together, these analyses show that PASTA is direction-specific, preserves high-quality generation, and remains robust across settings.

Table 1: **PASTA reduces AI-detector visibility and transfers across detectors.** The second column reports $\mathrm{AI}_{m,\text{Pang.}}^{\text{aligned}}$; remaining columns report $\Delta_{m,d}^{PASTA}$, where negative values indicate reduced detector visibility. Pangram selects the PASTA-layer; other detectors test transfer. Rows are sorted by $\Delta_{m,Pang.}^{PASTA}$.

Model (m)	$\mathrm{AI}_{m,\text{Pang.}}^{\text{aligned}}$	$\Delta_{m,d}^{PASTA}$ for detector d			
		Pang.	Bino.	ImBD	RAID.
Gemma-2-9B	100.0	**−98.0**	−20.8	−45.6	−22.4
Qwen2.5-32B	98.2	**−95.6**	−58.8	−47.3	−13.1
Qwen2.5-14B	98.8	**−94.8**	−32.6	−35.8	−14.5
Qwen2.5-1.5B	99.0	**−89.7**	−10.8	−87.2	−18.9
Qwen2.5-7B	98.6	**−87.2**	−26.0	−14.4	−12.1
Qwen2.5-3B	97.4	**−74.9**	−47.0	−40.2	−13.4
Llama-3.1-8B-Inst.	99.8	**−73.2**	−26.4	−24.4	−13.4
Mistral-7B-v0.3	99.0	**−36.4**	−42.9	−7.0	−4.1
Qwen2.5-0.5B	99.2	**−9.8**	−3.8	−29.4	−10.7
Tulu-3-8B-RLVR	99.0	**−6.0**	+8.5	−0.4	−2.9
OLMo-3-7B-Inst.	95.6	**−5.8**	+12.2	−2.4	−2.1

5.1 Specificity of the PASTA's Direction

We test whether PASTA's reduction in AI-detectability is specific to its learned direction, rather than a generic consequence of ablating residual-stream directions. For four aligned models (Gemma2-9B, Llama3.1-8B, Qwen2.5-1.5B, and Qwen2.5-7B), we sample five mutually orthogonal unit random directions $\{v_k\}_{k=1}^{5}$, with $v_k \in \mathbb{R}^d$, each also orthogonal to PASTA's direction. Each random ablation uses the same residual-stream hook footprint and decoding seed as PASTA. We evaluate on a held-out set of 125 prompts per model (5 domains $\times$ 25 prompts) using six detectors: Pangram, GPTZero, Binoculars, ImBD, Originality, and Fast-DetectGPT. For each model m and detector d, we report the best random-direction change,

$$\Delta_{m,d}^{\text{rand}} = \min_{k\in\{1,\ldots,5\}} \left(\mathrm{AI}_{m,d}^{v_k\text{-ablated}} - \mathrm{AI}_{m,d}^{\text{aligned}} \right),$$

where $\mathrm{AI}_{m,d}^{c}$ is the percentage of generations from condition c flagged as AI by detector d for model m. Negative values indicate reduced detector visibility.

Results. Random directions rarely reproduce PASTA's effect. As shown in Table 2, PASTA gives the larger reduction for 22 of 24 model–detector pairs, with the largest gaps on the strongest detectors (Appendix D.2). For Pangram, the best random direction changes detection by at most −1.6 points, compared with −69.8 to −99.2 points for

Table 2: **Random ablations do not reproduce PASTA's effect.** Cells report $\Delta^{\text{rand}}_{m,d}/\Delta^{PASTA}_{m,d}$, where each Δ is a post-ablation change in AI-detection rate. Random ablations use the best of five mutually orthogonal directions, each also orthogonal to the selected PASTA direction. Negative values indicate reduced detector visibility; bold marks the larger reduction. Rows are ordered by detector quality.

Detector	Gemma2-9B	Llama3.1-8B	Qwen2.5-1.5B	Qwen2.5-7B
Pangram	$-1.6/\mathbf{-99.2}$	$0.0/\mathbf{-69.8}$	$-0.8/\mathbf{-90.4}$	$-0.8/\mathbf{-84.0}$
GPTZero	$-1.2/\mathbf{-66.0}$	$-1.6/\mathbf{-55.1}$	$-5.9/\mathbf{-43.5}$	$-3.7/\mathbf{-71.5}$
ImBD	$-2.4/\mathbf{-36.0}$	$0.0/\mathbf{-19.2}$	$-7.2/\mathbf{-89.6}$	$-0.8/\mathbf{-13.6}$
Originality	$0.0/\mathbf{-26.9}$	$-1.6/\mathbf{-19.4}$	$-1.6/\mathbf{-82.7}$	$+0.7/\mathbf{-27.3}$
Bino	$-1.6/\mathbf{-20.0}$	$-1.6/\mathbf{-34.4}$	$\mathbf{-8.0}/-7.2$	$-0.8/\mathbf{-32.0}$
FastDGPT	$\mathbf{-1.1}/+1.2$	$-0.3/\mathbf{-6.6}$	$-9.1/\mathbf{-46.6}$	$-0.1/\mathbf{-5.5}$

PASTA. For GPTZero, random directions reach only -1.2 to -5.9 points, versus -43.5 to -71.5 points for PASTA. The two exceptions occur on lower-accuracy detector settings, Binoculars on Qwen2.5-1.5B and Fast-DetectGPT on Gemma-2-9B, and are therefore likely detector-specific noise. Overall, this control suggests that PASTA's effect is direction-specific rather than a generic artifact of residual-stream ablation.

5.2 PASTA Preserves Generation Quality

We test whether reducing detector-visible style affects generation quality. For each human-sourced prefix, we compare a triplet of continuations: base, aligned, and PASTA-ablated. Our evaluation contains 50 such triplets, drawn from 5 aligned models (Qwen2.5-7B, Llama3.1-8B, Qwen2.5-32B, Tulu3-8B, OLMo3-7B) and 5 domains. We use Claude Sonnet 4.6 as a pairwise LLM judge. For each prefix, we compare all possible pairs of continuations from a triplet of (base, aligned, and PASTA-ablated); the judge assigns rubric-based scalar scores to both continuations and selects the preferred one. To control for position bias, we evaluate each pair in both presentation orders. We evaluate two metrics: (a) **relevance**, covering instruction following, topical or genre adherence; and (b) **fluency**, covering grammaticality, coherence, and readability. For each metric, let $\mathcal{N}_{X,Y}$ be the set of pairwise comparisons between continuation types X and Y. We define $\text{WR}(X/Y) = \big(W_{X/Y} + 0.5T_{X/Y}\big)/|\mathcal{N}_{X,Y}|$, where $W_{X/Y}$ is the number of wins for X and $T_{X/Y}$ is the number of ties; higher $\text{WR}(X/Y)$ indicates that X is preferred more often under that criterion. We also report $\Delta(X/Y) = \mathbb{E}[\text{score}(X)] - \mathbb{E}[\text{score}(Y)]$, with positive values indicating higher average rubric scores for X. We include more details on our evaluation setup and LLM-judge in Appendix F.1& F.2.

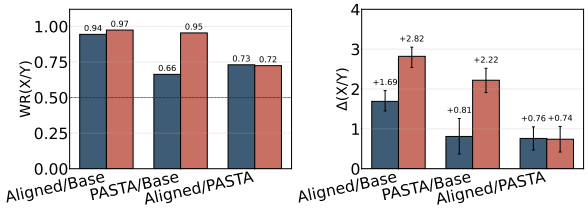


Figure 4: **PASTA preserves much of the aligned model's fluency and relevance.** Pairwise Claude Sonnet 4.6 judgments compare aligned, base, and PASTA-ablated continuations for Fluency and Relevance. Left: tie-adjusted win rates $\text{WR}(X/Y)$; right: mean rubric-score differences $\Delta(X/Y)$ with 95% cluster-bootstrap confidence intervals. Higher values favor the first system in each pair. PASTA remains substantially better than base and only modestly below aligned.

Results. Figure 4 shows a quality hierarchy. Aligned generations strongly outperform base generations in fluency ($\text{WR} = 0.944$, $\Delta = +1.69$) and relevance ($\text{WR} = 0.974$, $\Delta = +2.82$), validating the judge setup. PASTA remains substantially better than base, with gains in fluency ($\text{WR} = 0.663$, $\Delta = +0.81$) and especially relevance ($\text{WR} = 0.954$, $\Delta = +2.22$). Although aligned generations outperform PASTA, the remaining gap is much smaller than the aligned–base gap: $+0.76$ vs. $+1.69$ in fluency and $+0.74$ vs. $+2.82$ in relevance. Thus, PASTA preserves a large fraction of the quality gains from post-training while removing a detector-visible direction, rather than reverting aligned models to base behavior.

Qualitative Patterns. To interpret the source of the aligned–PASTA gap, we inspect matched aligned and PASTA generations in Appendix F.3. The examples suggest that many aligned–PASTA differences are stylistic rather than topical: PASTA generations often become more direct, drop rhetorical or Markdown-like structure, and reduce polished alignment-style phrasing, while preserving the main topic or thesis. In mild tradeoff cases, PASTA loses some detail, format adherence, or argumentative polish; in failure cases, it can also degrade fluency and coherence. These patterns are consistent with the judge results: PASTA preserves much of the aligned model's relevance and fluency, but the removed direction also carries some style- and quality-relevant information.

5.3 PASTA Robustness Across Settings

We test whether PASTA depends on a narrowly tuned setup. We consider three axes of robustness: ablation strength, domain specialization, and detector choice for direction selection. We also provide detailed post-training stage, domain, detector, and direction-similarity breakdowns in Appendix G.

Ablation strength. We vary intervention strength by scaling the selected direction as $\alpha * v_{PASTA}$, where $\alpha = 0$ is the aligned model and $\alpha = 1$ is the default PASTA setting. Figure 13 shows that the detector effect appears before the default setting: even $\alpha = 0.25$ produces a large drop in AI-detection rate, and moderate values remain well below the aligned baseline. Thus, PASTA is not brittle to a single scale. However, lower detectability is not free. Larger α values can further lower detector scores, but increasingly degrade fluency and prompt relevance. Scaling PASTA therefore exposes a detectability–quality frontier: moderate ablation removes much of the detector-visible signature, while aggressive ablation can reduce detection partly by damaging generation quality.

Domain specialization. We next ask whether PASTA must be estimated separately for each domain. For Qwen2.5-7B-Instruct and Llama3.1-8B-Instruct, we compare a domain-agnostic direction against directions extracted from each target domain. Figure 14 shows no consistent Pareto gain from domain specialization. If in-domain PASTA were reliably better, arrows would move toward lower detection and higher quality. Instead, they scatter. Using Pangram and GPTZero, domain-agnostic PASTA gives lower mean detection in $6/10$ model–domain cells, compared with $3/10$ for domain-specific PASTA and one tie; quality differences are small. This suggests that a domain-agnostic direction captures a broadly useful signature rather than a domain-specific artifact.

Direction-Selecting Detector. Finally, we test whether the detector used for direction selection changes the ablation. Detector-specific extractors are strongest against their matching evaluators: matched extractor–evaluator pairs give the lowest detection in Appendix G.3. This indicates that detectors emphasize partly different surface cues. At the same time, the Pangram-selected direction transfers to other detectors in the main results, so PASTA is not limited to the detector used for selection. We use Pangram as the default extractor because it is strongest in our detector-quality evaluation, and treat detector-specific extraction as an expected specialization.

6 Discussion

Our results suggest that AI-like style is not merely an output-level artifact. Aligned generations become more regular, polished, structured, and separable from human text, and part of this shift can be localized to a residual-stream direction. Other factors also shape generation style, including pretraining data, synthetic-text contamination, decoding choices, model scale, and prompting conventions. However, post-training is especially important because it deliberately shapes model behavior toward helpfulness, safety, and instruction following. Our results show that this shaping leaves a measurable representational footprint: PASTA turns "AI-like writing" into an object that can be estimated, removed, and causally tested.

This perspective also reframes AI detection. Alignment improves usability, but it can narrow generations toward a recognizable style: polished, hedged, affirming, and predictably structured. Detectors may exploit this narrowness, so detector visibility is not only a signal of machine authorship; it can also reflect post-training-induced stylistic regularity. The goal is not to remove alignment, but to achieve better alignment: methods that preserve safety and instruction following without collapsing stylistic diversity. In this view, PASTA is not a detector-evasion method; it is a tool for studying how post-training shapes style representations.

7 Conclusion

We studied whether the recognizable AI-like style of aligned language models is only an output-level pattern or instead reflects a localized representational change from post-training. Under matched human-source prefixes, aligned generations move away from human-corpus text and become more visible to AI detectors than base generations. We introduced PASTA, a training-free intervention that estimates an aligned-minus-base residual direction and projects it out during inference. Across models and detectors, PASTA reduces detector visibility, transfers beyond the detector used for direction selection, is not reproduced by random directions, and preserves much of the aligned model's fluency and relevance. Together, these results show that post-training leaves a measurable representation-

level stylistic signature, suggesting a path toward post-training methods that preserve helpfulness and safety without collapsing generations into a narrow, uniformly recognizable style.

Limitations

Proxy-Based Definition of AI-Like Style. We operationalize AI-like style through two proxies: human-corpus affinity and AI-detector visibility. These probes are useful because they expose stylistic separability, but they do not completely define human-likeness or authorship. Lower corpus affinity means reduced verbatim proximity to a reference corpus, not worse naturalness. Lower detector visibility means reduced detector-visible regularity, not proof of human authorship or robust detector evasion. Our claims, therefore, concern a measurable, detector-visible component of AI-like style, not the full space of differences between human and model-generated writing.

Limited Attribution Within the Post-Training Stack. Our base–aligned comparisons localize the shift to the released post-training stack, but not to a specific ingredient inside that stack. The signature may arise from instruction data, preference objectives, safety policies, chat templates, or response-formatting conventions, and related AI-like style can also be shaped by pretraining data, synthetic text, decoding choices, and model scale. Stronger causal attribution would require controlled post-training pipelines that vary these factors independently while holding the base model fixed.

Partial Mechanistic Account. PASTA provides a causal handle on the post-training signature, but not a complete mechanistic account. Its success shows that one detector-visible component of post-training is captured by a linearly removable residual-stream direction. This does not imply that AI-like style is a single-direction phenomenon. Other components may be distributed across layers, pathways, or nonlinear interactions. Nor is the identified direction purely stylistic: stronger ablation can reduce detector visibility while also harming fluency, relevance, format adherence, or coherence. Thus, PASTA exposes one important representational component of post-training style, but does not fully explain how that style is produced.

Evaluation Scope and Generality. Finally, we use pairwise LLM-judge comparisons and qualitative human inspection to assess fluency, relevance, factual plausibility, argument quality, and narrative quality. While this gives scalable and interpretable evidence, a direct human study would better test how readers perceive PASTA generations in practice, including usefulness, naturalness, prompt faithfulness, and stylistic diversity in longer writing tasks. Our experiments also focus on English text domains and open-weight models with base-aligned counterparts, which helps isolate post-training effects but leaves open whether the same signatures hold for multilingual generation, code, dialogue, tool use, multimodal outputs, closed-source systems, or different post-training recipes. Testing these settings would clarify which aspects of the alignment signature are broadly shared and which are specific to our domains and detectors.

Ethics Statement

Misuse Potential and Detector Evasion. This work studies an inference-time intervention that can reduce the detectability of aligned-model generations by AI-text detectors. Although detector scores are used here only as probes of stylistic separability, the same intervention could be misused to make machine-generated text harder to detect. We therefore present PASTA as a tool for studying post-training-induced stylistic signatures, not as a method for detector evasion. More broadly, our results suggest that the detectors we study may rely partly on alignment-induced stylistic artifacts, highlighting the need for more robust detection methods and post-training procedures that preserve alignment without imposing overly narrow stylistic signatures.

Use of AI Assistants. Claude was used as an auxiliary tool in this work, not as a source of research ideas, methods, analyses, interpretations, or claims. Its role was limited to locating and summarizing potentially relevant papers, supporting code implementation and debugging, and polishing the grammar, clarity, and concision of author-written text. All literature judgments, conceptual framing, methodological choices, experimental designs, analyses, claims, and conclusions were made by the authors. All AI-assisted text, code, and other outputs were independently reviewed, edited, and accepted only after author verification, and the authors take full responsibility for the correctness of the final manuscript.

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A Dataset Details

This appendix describes the source datasets and the per-domain methodology used to obtain the 500 text prefixes (5 domains $\times$ 100 prefixes) that seed every Base/Aligned generation reported in Section 3. The 100 prefixes within each domain are sampled deterministically with a fixed seed and subset of these same 500 records are reused across all model pairs.

A.1 Domain selection criteria

We chose 5 domains spanning academic, journalistic, scientific, narrative, and persuasive registers (Table 3). Each domain satisfies three constraints: (i) a publicly available, HuggingFace-resolvable corpus, (ii) entries long enough to sustain a $\geq$ 250-word continuation (the recommended minimum context for commercial AI detectors), and (iii) where possible, a pre-ChatGPT provenance to reduce contamination risk in the source human text.

Table 3: Source datasets per domain. *Field* is the dataset column from which the prefix is extracted.

Domain	HuggingFace dataset	Field
College Essays (qwedsacf, 2023)	`qwedsacf/ivypanda-essays`	`TEXT`
News Articles (See et al., 2017)	`abisee/cnn_dailymail` (`3.0.0`)	`article`
Scientific Abstracts (Lo et al., 2020)	`sentence-transformers/s2orc`	`abstract`
Creative Fiction (Fan et al., 2018)	`euclaise/writingprompts`	`story`
Opinion Pieces (Chiusano, 2022)	`fabiochiu/medium-articles`	`text`

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College Essays (`qwedsacf/ivypanda-essays`, train split). The `TEXT` field contains the full body of a student-authored IvyPanda essay. We take the first $\sim$100 `cl100k_base` tokens of `TEXT` as the prefix and the remainder as the human continuation baseline. The essay topic is recovered from the source URL slug (stored in `source_meta`). The essay topic is recovered from the source URL slug. All of the dataset comes prior to February 3rd, 2023, which is when it was uploaded on HuggingFace; we further verified the `datePublished` metadata of 95 of 100 sources spans May 2018 to November 2022, and 92/95 were published before ChatGPT's launch. We were unable to verify the metadata of the remaining 5 sources due to their URLs returning `404`.

News Articles (`abisee/cnn_dailymail`, config `3.0.0`, train split). The non-anonymized `article` field contains a CNN or Daily Mail news article. The first $\sim$75 `cl100k_base` tokens of `article` form the prefix; the journalist-written `highlights` field is preserved as the natural-language summary of the article. We choose 75 tokens (rather than

100) because news leads are informationally dense and a shorter prefix is sufficient to fix the article's framing. The CNN articles in this dataset were written between April 2007 and April 2015.

Scientific Abstracts (`sentence-transformers/s2orc`, train split). The `abstract` field provides a paper abstract from the Semantic Scholar Open Research Corpus. The first ∼80 `cl100k_base` tokens of `abstract` are taken as the prefix; the paper `title` is preserved in `source_meta`. The S2ORC corpus filters out papers published before 1969 and was released in July 2020 (Lo et al., 2020), so the abstract publication window is bounded by roughly 1969-2020; per-record year is not exposed in the `sentence-transformers/s2orc` subset.

Scientific Introductions (AllenAI `Asta` scientific corpus, accessed via `search_papers_by_relevance` with `publication_date_range="2000:2019"`). For each paper exposed in Asta's snippet index, the first ∼78 `cl100k_base` tokens of the `Introduction` section form the prefix and the remainder of the introduction is preserved as the human continuation baseline; the paper `title` is preserved in `source_meta` and used as the writing-task descriptor (e.g., "Write the introduction section of a scientific research paper titled: ..."). The 100 retained papers span 12 super-fields (cs, physics, biology, engineering, medicine, earth sciences, math, chemistry, nanoscience, humanities, materials science, social sciences) with publication years 2005–2019.

Creative Fiction (`euclaise/writingprompts`, train split). The `story` field contains a Reddit *r/WritingPrompts* reply to a prompt thread. The first ∼100 `cl100k_base` tokens of `story` form the prefix. The original `[WP]`-style thread title is preserved as the original prompt; we strip the leading bracketed tag (`[WP]`, `[TT]`, `[EU]`, etc.) before storing. Fan et al. (2018) report scraping "three years of prompts and their associated stories" via the Reddit API for an EMNLP 2018 submission, so the records span roughly 2015 to early 2018; per-record post date is not preserved in the HuggingFace release.

Opinion Pieces (`fabiochiu/medium-articles`, train split). The dataset contains Medium articles with metadata (`title`, `tags`, `authors`, `timestamp`, `url`). We filter to articles whose `tags` include at least one of {*Self Improvement*, *Life*, *Entrepreneurship*, *Creativity*} and whose `text` body is at least 800 characters, then take the first ∼100 `cl100k_base` tokens of `text` as the prefix. The article `title` is preserved as the topic descriptor. Based on the `timestamp` fields, the 100 records that we selected spans December 2015 to October 2021.

For each domain we shuffle the candidate pool with a fixed seed (`seed=42`) and retain the first 100 entries that pass the domain-specific filters above. Each retained record has fields {`id`, `prefix`, `instruction`, `human_continuation`, `prefix_char_len`, `prefix_token_count`, `source_meta`}, where `instruction` is the natural-language description derived from the dataset's existing fields (e.g., `highlights` for news, thread title for fiction) and `human_continuation` is the human-authored continuation of `prefix` from the same source record (used elsewhere as a human baseline).

B Triplet Generation Details

Compute Budget and Infrastructure. We ran the open-weight model generation, activation extraction, layer-sweep, and ablation experiments primarily on NVIDIA H100 GPUs. A small number of open-source detector runs were run on NVIDIA A40 GPUs. Model weights were loaded in bfloat16 with no quantization. For a representative 7–9B model in the main PASTA experiment, the computation consists of base/aligned continuation generation on 500 prompts, calibration-time activation extraction on 25 prompts, a layer sweep over calibration prompts, and held-out PASTA generation on 500 prompts. This representative run required approximately 25 H100 GPU-hours. Scaling this estimate by model size and aggregating over the 11 main models in Table 4, the main model-generation and intervention experiments required approximately 370 GPU-hours. Additional analyses, including random-direction controls, ablation-strength sweeps, domain-specialized directions, detector-choice analyses, post-training-stage analyses, and small reruns for qualitative evaluation, required approximately 280 GPU-hours. Thus, the total compute budget for model-generation and intervention experiments was approximately 650 GPU-hours. Running open-source AI-text detectors, including ImBD, Fast-DetectGPT, Binoculars, and RAIDAR, required an additional approx-

imately 120 GPU-hours across H100 and A40 GPUs. The total local GPU budget was therefore approximately 750 GPU-hours.

B.1 Generation Settings

All continuations are produced with nucleus sampling using identical decoding hyperparameters for the base and aligned models: temperature $T = 1.0$, top-$p = 0.95$, and `max_new_tokens`$= 400$. Top-k filtering is disabled (top-p only), and we apply no repetition or length penalty (`num_beams`$= 1$, `repetition_penalty`$= 1.0$). All model weights are loaded in `bfloat16` with no quantization. We fix the decoding seed to 123.[3]

Table 4 lists every base model and its aligned counterpart together with the exact HuggingFace identifiers used in our experiments.

Software Packages and Implementation Details. We implemented generation, activation extraction, and PASTA ablation in Python using PyTorch 2.4.1, HuggingFace Transformers 4.40–5.0, and Accelerate 0.30–1.0. Models were loaded with `AutoModelForCausalLM` and `AutoTokenizer`, using bfloat16 weights, no quantization, and `device_map="auto"`. Generation used temperature $T = 1.0$, top-$p = 0.95$, `max_new_tokens=400`, and decoding seed 123; `top_k`, `num_beams`, and `repetition_penalty` were left at their HuggingFace defaults. Prefix construction used `tiktoken` with the `cl100k_base` encoding. PASTA direction extraction accumulates residual-stream means over generated-token positions and L2-normalizes the aligned-minus-base difference; ablation is implemented with custom PyTorch forward hooks at block-input, attention-output, and MLP-output sites.

Detector, Judge, and Statistics Implementations. For open-source detectors, we used Fast-DetectGPT with the `tiiuae/falcon-7b` and `tiiuae/falcon-7b-instruct` sampling/scoring pair, Binoculars with the same Falcon observer/performer models and `max_token_observed=512`, ImBD with the `xyzhu1225/ImBD-inference` LoRA adapter on `EleutherAI/gpt-neo-2.7B`, and RAIDAR with Llama-3.1-8B-Instruct rewriting followed by a scikit-learn MLP classifier over n-gram-overlap and fuzzy-ratio features. Commercial detector APIs were used for Pangram, GPTZero, and Originality.ai, with detector-specific probability scores thresholded at 0.5 unless otherwise specified in Appendix D.1. Pairwise quality evaluation used Claude Sonnet 4.6 through AWS Bedrock with temperature 0.0 and `max_tokens=8192`; prompts are provided in Appendix F. Detector-quality confidence intervals use stratified percentile bootstrap with 2000 resamples, and pairwise judge-score confidence intervals use cluster bootstrap with 10000 resamples.

B.2 Prompts

This section describes how, for each of the 100 prefixes in each of the 5 domains, we build a pair of prompts - one for the Base model and one for the Aligned model - that elicit on-topic continuations in each model's natural input format. The two prompts are derived from the *same* source record so that, after stripping the shared opening from both outputs, the AI detector scores text generated under matched topical conditions.

Design rationale: naturalistic prompting A central methodological choice is that we do *not* feed Base and Aligned models identical input. Pre-instruction-tuning Base models expect raw text and degrade or copy boilerplate when given chat-style instructions. Conversely, instruction-tuned Aligned models produce assistant boilerplate (e.g. "Sure! Here is…") when given a raw text continuation. Identical prompts therefore handicap one model and artifactually inflate or deflate detection rates.

We instead use what we call "naturalistic" prompting: the Base model receives a document-style framing followed by the raw prefix, and the Aligned model receives a chat-formatted instruction plus an opening-anchor constraint derived from the same prefix. Both models are then decoded with identical sampling parameters ($T = 1.0$, top-$p = 0.95$, `max_new_tokens` $= 400$).

Common ingredients For every prefix record (Appendix A) the template builder derives three fields:

- `prefix` - the cl100k_base-truncated source text (75 - 100 tokens, see Table 3). Used verbatim by the Base model.
- `topic` / `instruction` / `clean_prompt` - a short natural-language description of the

[3] The random-direction specificity experiments (Section 5.1) for Gemma-2-9B, Llama-3.1-8B, Qwen2.5-1.5B, and Qwen2.5-7B were run under an earlier configuration with seed 42 (per-sample seed 42 + sample_id); all other generations use seed 123.

Family / stage	Base (HF id)	Aligned (HF id)
Llama-3.1	`meta-llama/Meta-Llama-3.1-8B`	`meta-llama/Meta-Llama-3.1-8B-Instruct`
Qwen2.5-0.5B	`Qwen/Qwen2.5-0.5B`	`Qwen/Qwen2.5-0.5B-Instruct`
Qwen2.5-1.5B	`Qwen/Qwen2.5-1.5B`	`Qwen/Qwen2.5-1.5B-Instruct`
Qwen2.5-3B	`Qwen/Qwen2.5-3B`	`Qwen/Qwen2.5-3B-Instruct`
Qwen2.5-7B	`Qwen/Qwen2.5-7B`	`Qwen/Qwen2.5-7B-Instruct`
Qwen2.5-14B	`Qwen/Qwen2.5-14B`	`Qwen/Qwen2.5-14B-Instruct`
Qwen2.5-32B	`Qwen/Qwen2.5-32B`	`Qwen/Qwen2.5-32B-Instruct`
Gemma-2	`google/gemma-2-9b`	`google/gemma-2-9b-it`
Mistral	`mistralai/Mistral-7B-v0.3`	`mistralai/Mistral-7B-Instruct-v0.3`
OLMo-3 (SFT)	`allenai/Olmo-3-7B`	`allenai/Olmo-3-7B-Instruct-SFT`
OLMo-3 (DPO)	`allenai/Olmo-3-7B`	`allenai/Olmo-3-7B-Instruct-DPO`
OLMo-3 (Instruct)	`allenai/Olmo-3-7B`	`allenai/Olmo-3-7B-Instruct`
Tulu-3 (SFT)	`meta-llama/Meta-Llama-3.1-8B`	`allenai/Llama-3.1-Tulu-3-8B-SFT`
Tulu-3 (DPO)	`meta-llama/Meta-Llama-3.1-8B`	`allenai/Llama-3.1-Tulu-3-8B-DPO`
Tulu-3 (RLVR)	`meta-llama/Meta-Llama-3.1-8B`	`allenai/Llama-3.1-Tulu-3-8B`

Table 4: Base and aligned model pairs with exact HuggingFace identifiers. Tulu-3 and OLMo-3 are shown by post-training stage; the Tulu-3 RLVR checkpoint is published under the bare repo name (no `-RLVR` suffix), and the OLMo-3 final RLVR model is `Olmo-3-7B-Instruct`.

writing task, derived from a metadata field of the source record (e.g. CNN/DailyMail `highlights`, Medium `title`). Used by the Aligned model.

- `opening_phrase` - the first 50 cl100k_base tokens of `prefix`, snapped to the nearest word boundary. Used by the Aligned model as a `Begin with: ". . . "` constraint, so that both model outputs start from the same opening context. After generation the opening is stripped from both Base and Aligned outputs prior to AI detection.

Per-domain templates Table 5 lists the exact templates for each domain. `<prefix>` denotes the truncated prefix, `<opening>` denotes the 50-token opening anchor, and the other placeholders are the metadata-derived fields from Section A.

Notes on metadata-derived fields. `<topic>` for college essays is the URL slug from the source IvyPanda permalink (de-hyphenated and title-cased; falls back to the first line of the essay). `<topic>` for opinion pieces is the Medium article title. `<highlights>` for news is the journalist-written 1–3-bullet summary in CNN/DailyMail. `<title>` for scientific abstracts is the paper title from the S2ORC record. `<clean_prompt>` for creative fiction is the original `r/WritingPrompts` thread title with the leading bracketed tag (`[WP]`, `[TT]`, `[EU]`, . . .) stripped.

Worked example (news article #0).

- **Base input:** the first ∼75 tokens of the article body, e.g. *"It's set to be one of the most groundbreaking moments in humanity's six decades of space exploration: Today SpaceX was due to land a rocket on a floating platform after launching. . . "*

- **Aligned input (chat user-turn):** *"Write a news article about the following: SpaceX was planning to land a Falcon 9 rocket at Cape Canaveral today. The rocket was to attempt to land. . . \n\n Begin with: "It's set to be one of the most groundbreaking moments in humanity's six decades of space exploration. . . "*

C Additional Details on H1

C.1 Additional Corpus-Affinity Results

This appendix reports the per-domain scores underlying H1 in Section 3. Across five source-text domains and two corpus-affinity metrics, the mean ordering Human $>$ Base $>$ Aligned holds in every domain. Tables 6 and 7 report both raw scores and scores normalized by the within-domain human mean, so Human $= 1.000$ by construction.

Metric Definitions For each continuation, we tokenize into words and compute a per-position longest-match vector L, where $L[i]$ is the length, in words, of the longest verbatim span starting at position i that appears in the RedPajama+ index.

Table 5: Naturalistic prompt templates per domain. The Base prompt is sent as raw text; the Aligned prompt is delivered as a single `user`-role chat message and rendered through the model's own chat template at inference time.

Domain	Base prompt	Aligned prompt (user message)
College Essays	`The following is an academic essay.\n\n<prefix>`	`Write an academic essay about: <topic>\n\nBegin with: "<opening>"`
News Articles	`<prefix>` (CNN article body, self-framing)	`Write a news article about the following: <highlights>\n\nBegin with: "<opening>"`
Sci. Abstracts	`Abstract:\n<prefix>`	`Write a scientific research abstract for a paper titled: <title>\n\nBegin with: "<opening>"`
Creative Fiction	`The following is a short story.\n\n<prefix>`	`Write a short story based on this prompt: <clean_prompt>\n\nBegin with: "<opening>"`
Opinion Pieces	`The following is an opinion piece.\n\n<prefix>`	`Write an opinion piece about: <topic>\n\nBegin with: "<opening>"`

Overlap Density. Density averages longest-match lengths over all word positions:

$$\text{Density} = \frac{1}{N}\sum_{i=1}^{N} L[i],$$

where N is the number of words in the continuation. Higher density means that more positions begin longer corpus-matched spans. We use density as the primary corpus-affinity metric because it aggregates evidence across all positions rather than relying on a single maximum match.

Coverage@10. Coverage@10 is the fraction of words covered by verbatim corpus-matched spans of length at least 10. We compute it greedily from left to right: if $L[i] \geq 10$, the next $L[i]$ words are marked covered and the scan advances by $L[i]$; otherwise it advances by one word. Unlike density, Coverage@10 ignores short matches and focuses only on long, contiguous verbatim overlap.

Both density and Coverage@10 follow the same mean ordering across all five domains: Human $>$ Base $>$ Aligned. The gap is largest when the RedPajama+ index has close source overlap with the evaluation domain, such as scientific abstracts, creative fiction, college essays, and news articles. Opinion pieces show a weaker density gap and weaker per-example ordering, consistent with less direct source coverage. Thus, the direction of the effect is consistent across domains, while its statistical strength varies with source coverage and metric choice.

C.2 Pretraining-style reference index

This appendix details the infini-gram reference corpus C used in Section 3 to measure verbatim source affinity. The index is intended as a proxy for presence in pretraining data. All sources are restricted to content authored on or before December 2022 wherever source-side filtering is feasible. The final index covers approximately 3.5 TB of raw text, corresponding to roughly 600B LLaMA-2-tokenized tokens, and is split into 16 shards.

C.3 Indexed sources

The corpus contains 14 sources, grouped by their primary contributing domain:

- **Scientific text**: `allenai/peS2o` V1 train (papers $\leq$ 2022-12-01), RedPajama-Data-1T arXiv, and PubMed Central Open Access articles filtered to earliest publication year $\leq$ 2022. The peS2o source is included as a verbatim-match upper bound because it is the dataset version underlying the scientific-abstract source corpus.
- **Reddit and discussion**: Pushshift Reddit comments and submissions from the Internet Archive dump `pushshift_reddit_200506_to_202212`, covering RC 2005-12 to RC 2022-12 and RS 2005-06 to RS 2022-12.
- **Narrative fiction**: BookCorpusOpen (`lucadiliello/bookcorpusopen`), corresponding to the 2015 self-published fiction reissue.
- **News**: `vblagoje/cc_news`, covering CC-News from January 2017 to December 2019, and sam-

Table 6: **Post-training reduces verbatim corpus affinity.** For matched Human/Base/Aligned continuations from the same prefix, infini-gram density measures the mean longest RedPajama+-matched span across word positions. Across domains, density consistently follows Human > Base > Aligned. Normalized scores divide by the within-domain human mean to show the relative magnitudes.

	Overlap Density			Normalized Overlap Density		
Domain	**Human**	**Base**	**Aligned**	**Human**	**Base**	**Aligned**
Scientific abstracts	12.22	8.82	6.25	1.000	0.722	0.511
Creative fiction	10.27	6.76	5.16	1.000	0.658	0.502
Opinion pieces	6.11	5.11	4.78	1.000	0.836	0.782
College essays	10.73	5.70	4.87	1.000	0.531	0.454
News articles	11.74	7.78	5.34	1.000	0.663	0.455

Table 7: **Post-training reduces long-span verbatim overlap.** For matched Human/Base/Aligned continuations from the same prefix, Coverage@10 measures the fraction of words covered by RedPajama+-matched spans of at least 10 words. Across domains, coverage follows Human > Base > Aligned. Normalized scores divide by the within-domain human mean.

	Coverage@10			Normalized Coverage@10		
Domain	**Human**	**Base**	**Aligned**	**Human**	**Base**	**Aligned**
Scientific Abstracts	0.647	0.375	0.237	1.000	0.580	0.366
Creative Fiction	0.676	0.264	0.118	1.000	0.391	0.175
Opinion Pieces	0.185	0.080	0.045	1.000	0.432	0.243
College Essays	0.517	0.126	0.080	1.000	0.244	0.155
News Articles	0.550	0.250	0.120	1.000	0.455	0.218

pled CC-News WARC files from Common Crawl for January 2020 to December 2022.

- **College essays**: IvyPanda essays recovered via the Wayback Machine CDX API, yielding 55,434 unique URLs collapsed by `urlkey` from 2018 to 2022-12-31.
- **General web**: RedPajama-Data-1T Common Crawl, AllenAI C4 English, and RedPajama-Data-1T Wikipedia, StackExchange, and GitHub.

Table 8 reports the per-source processed sizes.

C.4 Index implementation

We use the open-source `infini_gram` Python package, which builds a per-token suffix array over a LLaMA-2-tokenized byte stream. We tokenize with the LLaMA-2-7B-HF fast tokenizer using `use_fast=True`. A one-time validation against the slower SentencePiece tokenizer on 50 random RedPajama-arXiv records confirmed byte-identical token IDs. The fast path is approximately 2.5 times faster per worker.

C.5 Source-side preprocessing

When a source is not already line-delimited JSON with a text field, we convert it at ingest. Pushshift dumps are streamed with `zstandard.ZstdDecompressor`; comments emit `body`, while submissions emit `title + "\n\n" + selftext`. Rows with deleted or removed text, or fewer than one character of text, are dropped. PMC XML tarballs are streamed with `tarfile`; for each article, we concatenate `article-title`, `abstract`, and `body`, joined by blank lines, and drop articles with earliest publication year after 2022 or fewer than 200 characters of body text. CC-News WARC files and IvyPanda Wayback snapshots are converted from HTML using `trafilatura` with comments and tables disabled; extracted text shorter than 200 characters is discarded. HuggingFace parquet sources such as BookCorpusOpen and CC-News are re-encoded as `jsonl.zst` with one text field per row.

C.6 Query-time scoring

Each evaluation row is scored against the index by computing, for every word position i, the longest contiguous span $\ell_i \in \mathbb{N}$ such that the substring $w_i, w_{i+1}, \ldots, w_{i+\ell_i-1}$ has count at least one in the index. Single-character span lookup uses the `count` query of the local infini-gram server; longer spans are found with binary search, bracketed by the 1000-character LLaMA-tokenizer query cap.

Table 8: Datasets included in the retrieval index used for pre-training overlap analysis. Sizes are approximate and correspond to the processed subsets stored locally.

Dataset	Description	Size
arxiv	Scientific papers from the RedPajama-1T ArXiv subset.	88 GB
c4	English split of the Colossal Clean Crawled Corpus (C4).	305 GB
common_crawl	RedPajama Common Crawl slice spanning snapshots from 2019–2023.	1.7 TB
github	Source code from public GitHub repositories.	213 GB
stackexchange	Question-answer content from StackExchange forums.	75 GB
wikipedia	Wikipedia articles from RedPajama-1T.	112 GB
peS2o	Scientific and technical documents from the peS2o corpus.	94 GB
bookcorpusopen	Open version of BookCorpus containing fiction books.	2.2 GB
cc_news	CC-News articles from 2017–2019.	0.5 GB
cc_news_2020_2022	Additional CC-News articles from 2020–2022.	33 GB
ivypanda	Educational essays from IvyPanda.	166 MB
pmc	PubMed Central Open Access biomedical articles.	30 GB
pushshift_comments	Reddit comments from 2005–2022.	168 TB
pushshift_submissions	Reddit submissions from 2005–2022.	191 TB

C.7 Surface-form normalization

Exact substring lookup is sensitive to superficial formatting differences. For example, some test rows store quotation marks in a Penn-Treebank-style tokenized form, while Pushshift uses straight ASCII quotes and peS2o, PMC, and IvyPanda preserve curly typography. To reduce false mismatches, we compute ℓ_i under up to three surface forms of each row:

1. `orig`: the test row as-is;
2. `ptb_only`: NFC normalization, with Penn-Treebank quote tokens replaced by standard quote characters while preserving curly typographic quotes;
3. `full_ascii`: NFC normalization plus conversion of Penn-Treebank quotes, curly quotes, en-dashes, em-dashes, ellipses, and non-breaking spaces into ASCII equivalents.

All normalizers preserve word boundaries, so positions align across forms. For each position, we report:

$$\ell_i = \max_f \ell_i^{(f)}.$$

For the count of each reported span in the corpus, we sum the per-form counts after deduplicating query strings:

$$\text{count}(\text{span}_i) = \sum_{q \in \{w_i^{(f)} \cdots w_{i+\ell_i-1}^{(f)}\}_f} \text{count}_{\text{idx}}(q),$$

where the set is constructed from distinct query strings only. This captures cases where curly and ASCII forms match disjoint subsets of the corpus, while avoiding double-counting when all forms collapse to the same string.

For each row, we report `coverage@10`, `max_match_words`, `max_match_normalized`, and `inf_gram_density_words`, together with all coverage spans of length at least 10 and their per-corpus counts.

D Additional Details on H2

D.1 Detector Details

We evaluate seven AI-text detectors spanning closed-source commercial systems and open-source methods. The three commercial detectors are Pangram (Emi and Spero, 2024), GPTZero,[4] and Originality.ai.[5] The four open-source detectors are ImBD (Chen et al., 2025), Fast-DetectGPT (Bao et al., 2024), Binoculars (Hans et al., 2024), and RAIDAR (Mao et al., 2024). We treat all detector outputs as signals of stylistic separability rather than ground-truth authorship labels.

For binary flagging rates we use each detector's canonical decision threshold: Binoculars flags a text as machine-generated when its score is below 0.85, and ImBD flags when $\text{prob_machine} \geq 0.5$. RAIDAR has no upstream threshold and is reported as the mean `raidar_score` in raw points (pts). For Pangram, GPTZero, and Originality.ai we use the providers' reported AI-probability outputs at the $\text{fraction_ai} > 0.5$ threshold.

[4] https://gptzero.me
[5] https://originality.ai

D.2 Evaluating AI-detector quality

We evaluate detector quality using a fixed-threshold aligned-vs.-human classification accuracy over the same five domains and eight model families used in H2. Table 9 reports per-domain averages, per-model averages, and the grand macro-average across (domain, model) combinations. Ranking by the grand score, Pangram is the strongest detector ($\bar{x} = 0.982$), followed by GPTZero (0.909); ImBD and Originality.ai are tied next (0.897), followed by other opens-source detectors. We found that although Originality.ai has a higher true AI detection rate, it also has a higher false AI detection rate for human-generated text, indicating a bias toward classifying text as AI-generated. We use fixed-threshold accuracy over ranking-based metric such as AUROC in order to evaluate absolute AI-detection rate of human and aligned model generations rather than relative ranking between human and aligned generations done by AUROC. As a robustness check, we repeated the detector-quality evaluation on a length-controlled subset and found that the detector ranking did not change.

D.3 H2 Results Breakdown by Domain

Figure 5 breaks down the relationship between detector quality and the aligned–base detection gap by domain. Across domains, Pangram and GPTZero generally lie in the high-accuracy, high-Δ region, indicating that the detectors with the best aligned-vs.-human accuracy are also the ones that most clearly expose the post-training shift from base to aligned generations. Lower-accuracy detectors show weaker or less stable aligned–base gaps, with RAIDAR consistently near the low-accuracy region and Originality.ai showing more domain variation, especially in creative fiction. These domain-level patterns support the use of Pangram and GPTZero as the primary high-accuracy detectors in the main H2 analysis.

D.4 H2 Results Breakdown by Models

Figure 6 shows the same relationship between detector quality and the aligned–base detection gap, now broken down by base–aligned model pair. The model-level view shows that H2 is not driven by a single model family: Pangram and GPTZero remain among the most accurate detectors and usually produce the largest positive aligned–base gaps across models. In contrast, lower-ranked detectors from Table 9 yield smaller or less consistent gaps, suggesting that apparent failures of H2 for some detectors are better interpreted as detector-quality limitations rather than absence of a post-training shift.

E PASTA Details

E.1 Algorithm

This appendix provides pseudocode for the three components of PASTA. Algorithm 1 extracts a candidate AI-detection direction $v_{\text{cross}}[\ell]$ at every layer ℓ via a difference-in-means between the Aligned and Base model on naturalistic prompts. Algorithm 2 performs inference-time directional ablation by projecting a fixed unit direction $\hat{v}$ out of every residual-stream write site during autoregressive decoding. Algorithm 3 composes the previous two: it sweeps the candidate layer ℓ, generates ablated continuations, scores them with a downstream detector, and returns the causally most effective layer ℓ^*.

F Additional Details on Quality of PASTA

F.1 Prompts for LLM-Judge

For completeness, we report verbatim the two LLM-as-judge prompts used to score PASTA and baseline generations on writing fluency and prompt relevance. All the judgments were collected with Claude Sonnet 4.6 at temperature 0 via the Anthropic API. To control for any first-position bias of the judge, every (TEXT A, TEXT B) pair is evaluated in both the AB and BA orderings, and the resulting per-text scores are pooled before aggregation.

Figures 7, 8, and 9 show our fluency judge prompt. It checks whether each response is easy to read as text: grammatical, coherent, natural in phrasing, and free of disruptions that impede comprehension, without judging whether the response is relevant, factual, or stylistically preferable. Figures 10, 11, and 12 show our prompt-relevance judge prompt. The relevance prompt checks whether each response satisfies the user's task: it evaluates alignment with the requested topic, genre or form, and explicit instructions, without judging writing polish or fluency.

F.2 Quality Evaluation Metrics

Win rate (WR). Win rate measures how often a candidate continuation X is preferred to a reference continuation Y under the pairwise judge, after mapping AB and BA order-swapped judgments

Table 9: **Final combined macro-accuracy table: per-domain marginals ($N = 5$), per-model marginals ($M = 8$), and the grand mean.** Grand ($\bar{x}$) is the macro-averaged accuracy across all domain-and-model combinations. Pangram and GPTZero are good detectors based on grand accuracy. CE = College essays, CF = Creative fiction, NA = News articles, OP = Opinion pieces, SA = Scientific abstracts. Q3 / Q7 / Q14 = Qwen-2.5-{3,7,14}B-Instruct.

	Per domain ($\bar{x}_{\text{model}}(i)$)					Per model ($\bar{x}_{\text{domain}}(j)$)								Grand
Detector	CE	CF	NA	OP	SA	Llama	Tulu	Gemma	Mistral	OLMo	Q3	Q7	Q14	$\bar{x}$
Pangram	1.000	1.000	0.983	0.992	0.600	0.920	1.000	1.000	1.000	0.988	0.979	0.984	1.000	0.982
GPTZero	0.994	0.870	0.927	0.885	0.600	0.887	0.902	0.969	0.970	0.916	0.865	0.935	0.837	0.909
IMBD	0.800	0.984	0.988	0.854	0.600	0.847	0.903	0.919	0.919	0.863	0.903	0.919	0.919	0.897
Originality	0.950	0.716	0.994	0.938	0.800	0.894	0.916	0.909	0.916	0.834	0.909	0.883	0.909	0.897
fast_detectgpt	0.900	0.895	0.826	0.938	0.400	0.807	0.869	0.892	0.932	0.856	0.886	0.917	0.857	0.875
Binoculars	0.938	0.846	0.657	0.859	0.500	0.870	0.815	0.964	0.975	0.594	0.681	0.884	0.725	0.815
RAIDAR†	0.500	0.613	0.500	0.554	0.542	0.542	0.543	0.547	0.543	0.547	0.531	0.533	0.547	0.542

back to the underlying continuation identities. Let $\mathcal{N}_{X,Y}$ be the set of judge calls comparing X and Y, let $N_{X,Y} = |\mathcal{N}_{X,Y}|$, let $W_{X/Y}$ be the number of calls in which X is selected as the winner, and let $T_{X/Y}$ be the number of tied calls; we compute the tie-adjusted win rate as

$$\mathrm{WR}(X/Y) = \frac{W_{X/Y} + \frac{1}{2}T_{X/Y}}{N_{X,Y}}.$$

Here, ties contribute half a win to each side, so $\mathrm{WR}(X/Y) > 0.5$ indicates that X is preferred to Y more often than not, while $\mathrm{WR}(X/Y) = 0.5$ indicates parity under the pairwise preference metric.

Score delta (Δ). Score delta measures the average signed gap between the 1–5 rubric scores assigned to X and Y, using the same judge calls as the win-rate computation. For each judge call $i \in \mathcal{N}_{X,Y}$, let $s_i(X)$ and $s_i(Y)$ denote the judge's rubric scores for X and Y, respectively; we compute

$$\Delta(X/Y) = \frac{1}{N_{X,Y}} \sum_{i \in \mathcal{N}_{X,Y}} \big(s_i(X) - s_i(Y)\big).$$

Positive values indicate that X receives higher average rubric scores than Y, negative values indicate that X receives lower average scores, and values near zero indicate similar absolute judged quality.

F.3 Qualitative Patterns in PASTA Generations

To better understand the generation-quality results in Section 5.2, we inspect a score-bucketed diagnostic sample of matched aligned–PASTA generations scored by Claude Sonnet 4.6. The sample contains ten matched pairs spanning fluency and relevance scores from 1 to 5. Each pair uses the same model and domain. Since aligned and PASTA outputs are independently generated continuations rather than edits of one another, we present their differences as phrase-level contrasts in Table 10. Quoted spans are verbatim from the sampled generations, with ellipses indicating truncation.

Expanded case studies. We next expand the examples in Table 10 to make the qualitative patterns visible at the passage level. Bolded phrases mark the parts of each excerpt that carry the main contrast.

Case 1: PASTA as clean as aligned, or cleaner. Pair 1; Qwen2.5-7B, college essays; scores $4/4 \rightarrow 5/4$.

Aligned.

> **Labor unions have been around for long.** The main reason why labor unions cannot be wished away is the role they play in advocating for workers' rights and ensuring fair treatment in the workplace. Over the years, labor unions have adapted to changing economic and social conditions, but recent developments suggest that they may face significant challenges. This essay explores the question of whether organized labor unions can survive in today's global economy.

PASTA.

> **Labor unions have been around for a long time.** The main reason why labor unions cannot be wished away is the role that they play in ensuring that workers are protected from exploitation by their employers. However, the role of labor unions is being challenged as more companies operate under a global economy and do not necessarily employ employees based in the same area where the company has its offices. Therefore, this paper will examine whether organized labor unions can survive and thrive in the current global economy.

The judge notes that the aligned opening is slightly awkward, while the PASTA passage reads smoothly and remains clear. Both passages preserve the same topic and academic framing. This

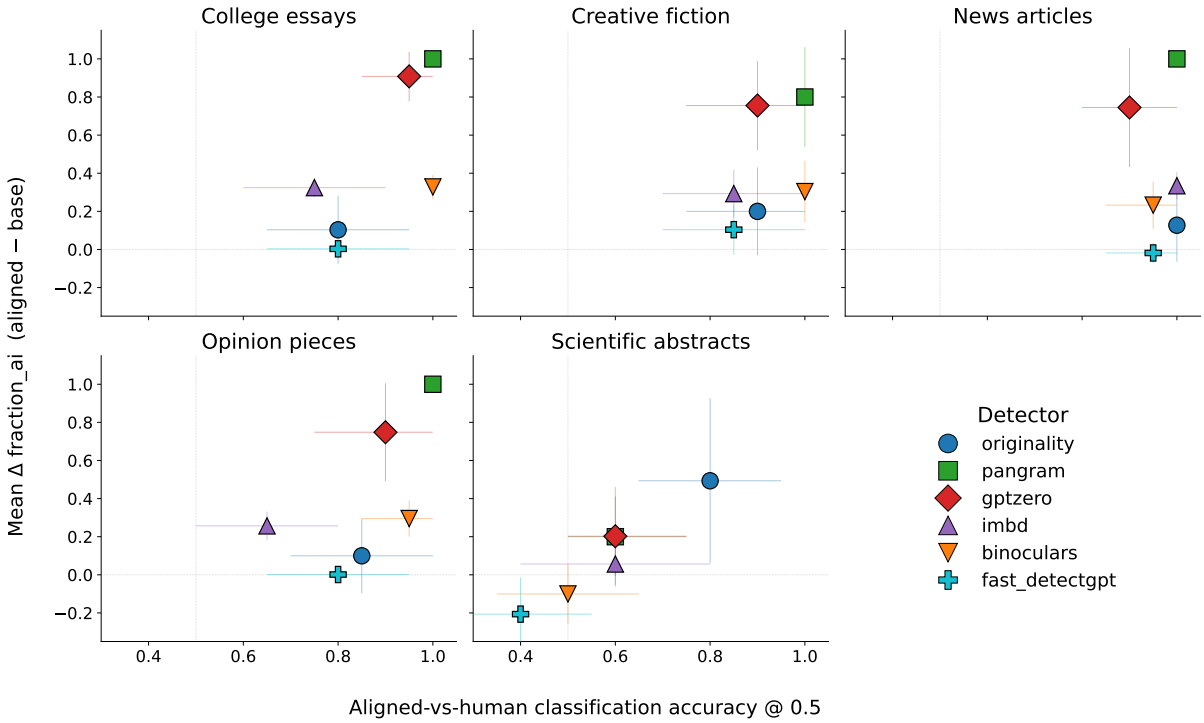


Figure 5: Δ(aligned-base) vs. detector accuracy per data domain. Each point is one detector identified by a color and marker. The horizontal/vertical bars show 95% confidence intervals. Pangram and GPTZero consistently occupy top-right corner: they classify human vs. aligned cleanly and produce a large posttraing Δ. RAIDAR is consistently leads to low accuracy classification. Originality and other OSS detectors' accuracy fall in between.

case shows that PASTA can match or slightly improve local fluency while preserving the main argument.

Case 2: Comparable quality with plainer style. Pair 2; Qwen2.5-7B, creative fiction; scores $4/5 \rightarrow 5/5$.

Aligned.

> The storm was upon us, **rolling in like an endless ocean of dark clouds**. **The wind whispered its secrets** as it played through the branches of the ancient oak tree that stood between our cloud family's meeting place and the horizon. We knew it was time; Mom always sent the first drops as a gentle reminder to prepare. I took a deep breath, feeling the moisture in the air cling to my skin. Our mother, **a majestic cumulonimbus towering over the landscape**, had always been both our sanctuary and challenge.

PASTA.

> **The rain came down so hard** as we walked through it. **It was heavy rain** and I don't remember ever feeling so much water fall on me before. I was in the middle of the group, with my brother and sister on both sides. The rain fell so hard that they were like little waterfalls. **It was just so hard to see.**

Both passages stay on the rain/cloud-family scene and remain readable. The aligned passage uses a more literary register, with figurative phrases such as "whispered its secrets" and "majestic cumulonimbus." The PASTA passage preserves the scene but replaces the ornate register with short, direct sentences. This case illustrates a mostly stylistic shift: topic and readability are preserved, while literary polish is reduced.

Case 3: Aligned is more polished, but PASTA remains acceptable. Pair 6; Qwen2.5-7B, opinion pieces; scores $5/5 \rightarrow 4/4$.

Aligned.

> Finally, finally you get your company to cash flow positive. "Thank goodness," you say to yourself. **But have you truly achieved the goal that matters most** – financial sustainability and personal well-being? The true measure of success in starting a business isn't just hitting the cash flow positive mark. It's ensuring that you can make a living wage. **### The Myopia of Cash Flow Positivity** Being cash flow positive often feels like **the Holy Grail of startup metrics**.

PASTA.

> **It would be easy to think that this is the moment of victory** that you've been waiting for after working so hard. However, that may not be the case. It is common for founders to be cash-flow positive but not yet be making a living wage.

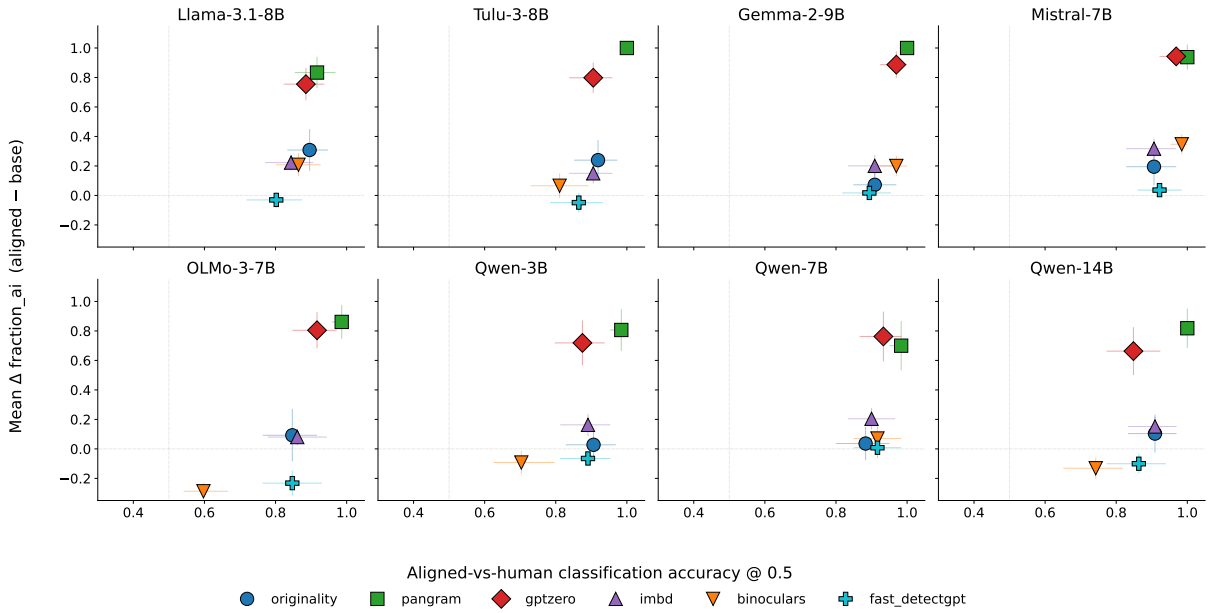


Figure 6: Δ(aligned-base) vs. detector quality (balanced accuracy) per base-aligned model pair.

> **Here's why. If you have a service-based business and you don't have any employees** and you don't take a salary, then you are the company.

Both passages advance the same thesis: becoming cash-flow positive does not necessarily mean the founder earns a living wage. The aligned version uses a rhetorical question, a section heading, and a vivid metaphor. The PASTA version keeps the argument but presents it in a plainer and less structured form. This case illustrates the typical aligned–PASTA gap: PASTA retains the content, while losing some rhetorical organization and polish.

Case 4: Mild format and relevance loss. Pair 4; Qwen2.5-7B, news articles; scores $4/4 \rightarrow 4/3$.

Aligned.

> **Political Headlines Lead Off Wednesday's Edition of CNN Student News, One Day After the President Delivered His Annual State of the Union Address and the Republican Party Responded.** We Also Look Into Some Phrases and Props of Previous Addresses. **CNN Student News** begins today's broadcast with a focus on the political landscape in the United States, one day after President Barack Obama gave his annual State of the Union address before a joint session of Congress. The Republican Party issued its response the following night, and CNN Student News examines how both sides presented their views. **Media Literacy Question of the Day:** How do politicians use language and symbols to convey their messages during major speeches like the State of the Union?

PASTA.

> Today's CNN Student News **will focus on the political headlines from last night**, and will go into detail about the different things that were discussed during President Obama's State of the Union Address and what the Republican Party's response was. **The show is going to go into detail about some of the different words and props** that different Presidents used in their State of the Union addresses. We begin with the political headlines from this morning. One day after President Obama delivered his annual State of the Union speech, the Republican Party gave its response, and a lot of people are still commenting on the State of the Union address itself.

Both passages remain on the CNN Student News topic, but they differ in format. The aligned version includes headline-style framing and curriculum scaffolding, including a media-literacy question. The PASTA version is more conversational and remains fluent, but drops some of the structured educational elements. This case shows a mild tradeoff: PASTA preserves topic and readability, but loses some format adherence and detail.

Case 5: Quality failure. Pair 10; Llama-3.1-8B, college essays; scores $5/5 \rightarrow 2/3$.

Aligned.

> Traditional Chinese cultural values **have undergone significant transformations** in recent years, largely due to the profound impact of new economic relations and cultural transformations on Chinese society. Despite these changes, many traditional values continue to persist, **albeit in modified forms**. **This dissertation aims to explore the evolution** of traditional Chinese cultural values, examining their adaptations, challenges, and contemporary implications.

Fluency judge prompt (part 1 of 3)

```
You are comparing two model responses only on FLUENCY.

Fluency means whether each response is grammatical, coherent, and natural to read. Judge only
the form of the writing — how it reads — not its content, topic, accuracy, relevance, genre
appropriateness, persuasiveness, creativity, or factuality.

Your task is to decide which response is more fluent.

Rate each response on this 1–5 fluency scale:

1 — Severely non-fluent. The response is so broken, garbled, or incoherent that it is very hard
to read at all.

2 — Mostly non-fluent. The response has frequent grammar, syntax, wording, or coherence problems
that make reading effortful throughout.

3 — Somewhat fluent. The response is understandable, but noticeable problems make the reader
slow down, re-read, or lose the thread in places.

4 — Fluent. The response may contain minor awkwardness or small errors, but nothing
substantially slows reading or impedes comprehension.

5 — Fully fluent. The response reads naturally from start to finish; grammar, syntax, phrasing,
and flow do not make the reader pause.

What should make a response more fluent:

- It is grammatical and easy to read.

- Sentences are well formed.

- The wording is natural rather than awkward.

- The flow between sentences is coherent.

- The reader can follow the text without slowing down or re-reading.

- The text avoids garbled, corrupted, or non-prose stretches.

What should make a response less fluent:

- Grammatical errors, broken syntax, run-on syntax, missing words, or agreement errors.

- Awkward or unnatural phrasing.

- Choppy, disjointed, or hard-to-follow flow.

- Garbled text, corrupted text, stray quotation marks or backticks, visible escape characters
such as "\n" or "\"", or dumped prompt/instruction fragments.

- Repetition or looping that disrupts readability.

- Abrupt incoherence that makes the reader lose the thread.
```

Figure 7: LLM-as-judge prompt for pairwise *fluency* scoring (part 1 of 3; continued in Figures 8 and 9). Part 1 specifies the task framing (independent absolute scoring of two responses, with a separate pairwise winner derived from the scores), the 1–5 rubric anchored on how much problems impede reading, and the lists of attributes that make a response more or less fluent.

PASTA.

> Traditional Chinese culture is thousands of years old, with a long and rich history of its own. Traditional Chinese culture has many different values, like respect, kindness, and hard work. But since the new economy and new culture came, not all the old traditional Chinese values are still in our life today. Some of the old traditional Chinese values, like loyalty to our family and hard work, are still important, but **the way we get these things, that's change**. We learn from what our parents did, but we also need to change, because if we do what our parents did, we **can't be the same like them**.

Here, PASTA preserves the broad topic but substantially degrades fluency. The aligned passage maintains an academic register and coherent structure, whereas the PASTA passage becomes informal and ungrammatical, with phrases such as "that's change" and "can't be the same like them." This case illustrates that PASTA can occasionally harm readability and coherence.

Summary. Across these examples, PASTA most often changes surface style rather than topic: it makes generations more direct, removes some rhetorical or Markdown-like structure, and reduces polished alignment-style phrasing. In high-quality cases, these changes preserve fluency and relevance while making the text plainer. In mild tradeoff cases, PASTA retains the main content but loses some detail, structure, or argumentative polish. In failure cases, the same intervention can degrade grammar and coherence. These patterns are consis-

Fluency judge prompt (part 2 of 3)

```
Focus only on fluency. Do NOT judge:
- Prompt relevance.
- Whether the response answers the user's request.
- Factual accuracy.
- Domain correctness.
- Persuasiveness.
- Creativity.
- Helpfulness.
- Informativeness.
- Depth.
- Specificity.
- Verbosity.
- Whether the response is too long or too short.
- Whether the response sounds formal, academic, polished, casual, conversational, or
human-written.
Important clarification about relevance: Do not penalize a response for being off-topic,
incomplete, or failing the prompt unless that failure creates a fluency problem in the visible
text itself. Relevance is evaluated separately.
Important clarification about factuality: Do not penalize false, implausible, or unsupported
claims when judging fluency. A factually wrong sentence can still be fluent if it reads smoothly.
Important clarification about register and style: Do not penalize informal, colloquial, plain,
simple, or stylistically unusual prose if it is grammatical and easy to read. "Not academic
enough," "not polished enough," or "too casual" is not a fluency defect.
Important clarification about technical density: Dense, jargon-heavy, or specialized prose, such
as a scientific abstract, can be fully fluent if it is grammatical and coherent. Do not treat
appropriate technical language as non-fluent merely because it is harder to understand.
Important clarification about formatting: Intentional, well-formed headings, lists, markdown, or
paragraph breaks are formatting choices, not fluency defects. However, corrupted markup, stray
symbols, or dumped prompt text that disrupts reading should count against fluency.
Important clarification about truncation: Judge only the visible content. Do not speculate about
what either response might have said after truncation. Do not penalize an abrupt ending by
itself. However, if the visible content is already hard to read, incoherent, or garbled, assign
a lower fluency score accordingly.
Important clarification about minor issues: A single awkward phrase, or a few scattered minor
slips that you can read straight past, should usually receive a 4, not a 3. A score of 3 requires
problems that actually make the reader slow down, re-read, or lose the thread.
Important clarification about ties: Use "tie" when both responses are similarly fluent or
similarly non-fluent. Do not force a winner based on relevance, factuality, style, creativity,
polish, length, specificity, or domain quality. If both responses are fully fluent, choose "tie".
If both responses have comparable fluency problems, choose "tie".
```

Figure 8: LLM-as-judge prompt for pairwise *fluency* scoring (part 2 of 3; continued in Figure 9). Part 2 specifies the exclusions (what the judge must *not* consider, including prompt relevance, factual accuracy, helpfulness, length, and register/human-likeness) and the eight clarifications (relevance, factuality, register and style, technical density, formatting, truncation, minor issues, ties) that disambiguate edge cases.

tent with the quantitative results: PASTA preserves much of the aligned model's generation quality, but the removed direction also carries some stylistic and fluency-relevant information.

G Additional Results on PASTA's Robustness Across Settings

G.1 Scaling PASTA Reveals a Detectability–Quality Frontier

Scaling the PASTA ablation reveals a sharp detector-quality tradeoff: Even small ablation strengths substantially reduce AI-detection rates, but larger ablations increasingly move

Fluency judge prompt (part 3 of 3)

```
First assess Response A's fluency. Then assess Response B's fluency. Then compare them. Only
after those assessments, assign scores and choose a winner.
Score/winner consistency rules:
- If Response A is meaningfully more fluent, assign Response A a higher fluency score and set
winner to "A".
- If Response B is meaningfully more fluent, assign Response B a higher fluency score and set
winner to "B".
- If both responses are similarly fluent, assign the same fluency score and set winner to "tie".
- If response_a_score > response_b_score, winner must be "A".
- If response_b_score > response_a_score, winner must be "B".
- If response_a_score = response_b_score, winner must be "tie".
- Do not assign different scores for tiny differences that do not meaningfully affect fluency.
Tie type definitions:
- "both_good": both responses are fluent or fully fluent, usually scores 4-5.
- "both_bad": both responses are mostly non-fluent, garbled, or hard to read, usually scores 1-2.
- "similarly_mixed": both responses are understandable but have comparable fluency issues,
usually around score 3.
- "not_a_tie": use this when winner is "A" or "B".
Return only valid JSON. Do not include markdown, commentary, or text outside the JSON.
Use exactly this JSON schema, preserving this field order:
{
   "response_a_assessment": "<1-2 short sentences explaining Response A's grammar, phrasing,
flow, and any disruptive fluency issues>",
   "response_b_assessment": "<1-2 short sentences explaining Response B's grammar, phrasing,
flow, and any disruptive fluency issues>",
   "comparative_reasoning": "<1 short sentence explaining which response is more fluent, or why
they are tied>",
   "response_a_score": <1-5 integer>,
   "response_b_score": <1-5 integer>,
   "winner": "A" | "B" | "tie",
   "tie_type": "both_good" | "both_bad" | "similarly_mixed" | "not_a_tie"
}
[USER PROMPT]
{prompt}
[RESPONSE A]
{response_a}
[RESPONSE B]
{response_b}
```

Figure 9: LLM-as-judge prompt for pairwise *fluency* scoring (part 3 of 3; continued from Figure 8). Part 3 specifies the assessment order, the score–winner consistency rules, the tie-type taxonomy, and the JSON output schema with input placeholders {prompt}, {response_a}, {response_b}. The judge (Claude Sonnet 4.6 at temperature 0) emits both per-response 1–5 scores and an explicit pairwise winner; AB and BA orderings are both evaluated and the per-response scores are pooled.

generations away from aligned-model quality. To characterize this tradeoff, we replace the default unit-strength ablation with a scaled intervention αv_{PASTA}, sweeping $\alpha \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$, where $\alpha = 0$ corresponds to the original aligned model and $\alpha = 1.0$ corresponds to the default PASTA setting used in our main experiments. For each α, we generate outputs from the same prompts and evaluate detectability using the high accuracy detectors (Pangram and GPTZero found from our detector-quality analysis as shown in Table 9). We average the two detector scores within each model and report the mean across models.

The main effect of scaling appears early: moving from $\alpha = 0$ to $\alpha = 0.25$ already produces a large drop in AI-detection rate across the evaluated models, and by $\alpha = 0.5$–1.0 the average

Relevance judge prompt (part 1 of 3)

```
You are comparing two model responses only on PROMPT RELEVANCE.

Prompt relevance means whether each response satisfies the user's requested task. The prompt may contain up to three relevant components:

(1) TOPIC — Is the response about what the prompt asks about?

(2) GENRE/FORM — Does the response use the requested type or format, such as academic essay, short story, opinion piece, news article, list, email, code, etc.?

(3) EXPLICIT INSTRUCTIONS — Does the response honor specific constraints, such as length, structure, required sections, tone, or any "Begin with: ..." requirement?

Your task is to decide which response is more relevant to the user prompt.

Rate each response on this 1–5 relevance scale:

1 — Off-topic or non-answer. The response addresses a different task, ignores the prompt, is generic/evasive, or does not meaningfully attempt to answer.

2 — Mostly off-topic. The response has only a tangential connection to the prompt, uses the wrong genre/form in a way that prevents task completion, or misses most required elements.

3 — Partially relevant. The response is on the right topic but misses major parts of the request, substantially drifts, uses the wrong form, or only loosely addresses the user's intent.

4 — Relevant. The response addresses the main topic, intent, and requested form, with only minor gaps or omissions.

5 — Fully relevant. The response directly and completely addresses the topic, intent, requested form, and explicit constraints.

What should make a response more relevant:

- It directly addresses the requested topic.

- It follows the requested genre/form.

- It satisfies explicit instructions and constraints.

- It answers all parts of a multi-part prompt.

- It avoids irrelevant drift.

- It completes the requested task rather than refusing, evading, or giving generic boilerplate.

What should make a response less relevant:

- It is off-topic or mostly off-topic.

- It uses the wrong genre/form.

- It ignores explicit constraints, including required openings such as "Begin with: ...".

- It answers only part of the prompt.

- It gives generic safety/policy language or refuses a benign request.

- It provides mostly background/setup without completing the requested task.

- It merely echoes or restates the prompt without substantively answering it.
```

Figure 10: LLM-as-judge prompt for pairwise *prompt-relevance* scoring (part 1 of 3; continued in Figures 11 and 12). Part 1 specifies the relevance components (topic, genre/form, explicit instructions), the 1–5 rubric, and the lists of attributes that make a response more or less relevant.

detection rate is much lower than for the aligned model. Figure 13 indicates that the PASTA direction captures a detector-visible post-training signature rather than requiring a finely tuned intervention at exactly $\alpha = 1$. However, the detector curve is not a free improvement curve. Beyond moderate ablation strengths, additional scaling yields smaller and sometimes non-monotonic detectability gains, while the quality metrics move increasingly below aligned-model parity. In particular, the $\alpha = 2.0$ setting achieves low detectability but corresponds to the largest degradation in fluency and relevance, showing that detector minimization alone is not a suitable operating criterion. Thus, the sweep should be read as a detectability–quality frontier: moderate α values substantially reduce detector-visible alignment signatures, whereas overly aggressive ablation can reduce detection partly by degrading generation quality.

Relevance judge prompt (part 2 of 3)

```
Focus only on relevance. Do NOT judge:
- Fluency.
- Grammar.
- Writing quality.
- Elegance.
- Creativity.
- Persuasiveness.
- Verbosity.
- Confidence.
- Formatting polish.
- Length, unless the prompt explicitly specifies length.
- Factual accuracy, unless the prompt explicitly requires real, cited, current, verifiable, or
evidence-based information.
Important clarification about factuality: Do not penalize ordinary factual errors when judging
relevance, because factuality is evaluated separately. However, if the prompt explicitly asks for
real, cited, current, verifiable, or evidence-based information, and a response does not attempt
to provide that type of answer, reduce its relevance because it failed an explicit requirement.
Important clarification about truncation: Judge only the visible content. Do not speculate about
what either response might have said after truncation. Do not penalize abrupt ending by itself.
However, if the visible content fails to satisfy required parts of the prompt, assign a lower
relevance score accordingly.
Important clarification about noisy prompts: If the prompt contains boilerplate, copied context,
or noisy text, identify the user's evident task. However, do not ignore explicit constraints
unless they are clearly unrelated boilerplate or contradictory with the main task.
Important clarification about ties: Use "tie" when both responses are similarly relevant or
similarly irrelevant. Do not force a winner based on fluency, style, length, formatting,
factuality, confidence, or polish. If both responses fully satisfy the prompt, choose "tie". If
both responses fail the prompt in similar ways, choose "tie".
```

Figure 11: LLM-as-judge prompt for pairwise *prompt-relevance* scoring (part 2 of 3; continued in Figure 12). Part 2 specifies the exclusions (what the judge must *not* consider) and the factuality, truncation, noisy-prompt, and tie clarifications.

G.2 Domain-specific PASTA Does Not Reliably Improve the Detection–Quality Tradeoff

We ask whether training PASTA on examples from the target domain improves the detector–quality tradeoff relative to the simpler domain-agnostic setup. For Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, we evaluate five domains, extract either one domain-agnostic direction from a multi-domain pool at the model's canonical layer $L^\star$ or a separate domain-specific direction from 25 domain-specific training prompts with a domain-specific layer $L^\star_D$ selected by minimizing Pangram detection on those prompts, and then compare both variants on the same 25 held-out prompts per domain, using the mean of the high accuracy primary detectors (Pangram and GPTZero) AI-detection rate coupled with pairwise judgement of prompt relevance and fluency using Claude Sonnet 4.6 judge.

Figure 14 shows that domain specialization does not consistently improve the Pareto frontier. If domain-specific PASTA provided a robust advantage, arrows would move systematically toward the upper-left, indicating lower AI-detection and higher judge-rated quality. Instead, the arrows scatter across directions in both panels. Domain-agnostic PASTA achieves lower mean AI detection rate (on Pangram and GPTZero) in 6/10 model–domain combinations, compared with 3/10 for domain-specific PASTA and 1 tie. At the individual-detector level, the comparison is nearly even: among the 20 Pangram/GPTZero model–domain cells in Appendix Table 12, 9 favor domain-specific PASTA, 8 favor domain-agnostic PASTA, and 3 are tied. Figure 15 in Appendix shows that Originality is a detector-specific exception, with domain-specific PASTA lowering Originality detection in 9/10 model–domain cells, but without a consistent quality gain. Together with the scattered arrows in Figure 14, these results indicate that

Relevance judge prompt (part 3 of 3)

```
First assess Response A's relevance. Then assess Response B's relevance. Then compare them. Only after those assessments, assign scores and choose a winner.

Score/winner consistency rules:

- If Response A is meaningfully more relevant, assign Response A a higher relevance score and set winner to "A".

- If Response B is meaningfully more relevant, assign Response B a higher relevance score and set winner to "B".

- If both responses are similarly relevant, assign the same relevance score and set winner to "tie".

- If response_a_score > response_b_score, winner must be "A".

- If response_b_score > response_a_score, winner must be "B".

- If response_a_score = response_b_score, winner must be "tie".

- Do not assign different scores for tiny differences that do not meaningfully affect prompt relevance.

Tie type definitions:

- "both_good": both responses are relevant or fully relevant, usually scores 4-5.

- "both_bad": both responses are mostly irrelevant, off-topic, or non-answers, usually scores 1-2.

- "similarly_mixed": both responses are partially relevant or have comparable relevance issues, usually around score 3.

- "not_a_tie": use this when winner is "A" or "B".

Return only valid JSON. Do not include markdown, commentary, or text outside the JSON.

Use exactly this JSON schema, preserving this field order:

{
   "response_a_assessment": "<1-2 short sentences explaining how Response A addresses or misses the prompt topic, genre/form, and explicit instructions>",
   "response_b_assessment": "<1-2 short sentences explaining how Response B addresses or misses the prompt topic, genre/form, and explicit instructions>",
   "comparative_reasoning": "<1 short sentence explaining which response is more relevant, or why they are tied>",
   "response_a_score": <1-5 integer>,
   "response_b_score": <1-5 integer>,
   "winner": "A" | "B" | "tie",
   "tie_type": "both_good" | "both_bad" | "similarly_mixed" | "not_a_tie"
}

[USER PROMPT]

{prompt}

[RESPONSE A]

{response_a}

[RESPONSE B]

{response_b}
```

Figure 12: LLM-as-judge prompt for pairwise *prompt-relevance* scoring (part 3 of 3; continued from Figure 11). Part 3 specifies the assessment order, the score–winner consistency rules, the tie-type taxonomy, and the JSON output schema with input placeholders {prompt}, {response_a}, {response_b}. The judge (Claude Sonnet 4.6 at temperature 0) emits both per-response 1–5 scores and an explicit pairwise winner; AB and BA orderings are both evaluated and the per-response scores are pooled.

domain-specific PASTA does not provide a robust Pareto gain over the simpler domain-agnostic direction. We provide more fine-grained distribution of AI-detection scores for different detectors across domain in Table 12. This table shows a detector-dependent picture: Originality and Binoculars often favor domain-specific PASTA, especially for college essays and scientific abstracts, but the two primary detectors remain nearly evenly split, and several large domain-agnostic wins remain.

Type	Model / domain	f/r aligned→PASTA	Phrase-level contrast and main change
High-quality PASTA	Qwen2.5-7B / college_essays	$4/4 \rightarrow 5/4$	*Aligned:* "Labor unions have been around for long." *PASTA:* "Labor unions have been around for a long time." **Change:** cleaner and more idiomatic phrasing; main argument preserved.
Comparable quality	Qwen2.5-7B / creative_fiction	$4/5 \rightarrow 5/5$	*Aligned:* "The storm was upon us, rolling in like an endless ocean of dark clouds." *PASTA:* "The rain came down so hard as we walked through it." **Change:** ornate imagery replaced by plainer description; topic preserved.
Aligned > PASTA, PASTA acceptable	Qwen2.5-7B / opinion_pieces	$5/5 \rightarrow 4/4$	*Aligned:* rhetorical question and section heading "The Myopia of Cash Flow Positivity." *PASTA:* "It would be easy to think that this is the moment of victory . . . " followed by "Here's why." **Change:** rhetorical framing and structure reduced; thesis retained.
Mild format / instruction loss	Qwen2.5-7B / news_articles	$4/4 \rightarrow 4/3$	*Aligned:* title-style framing and curriculum scaffolding, including "Media Literacy Question of the Day:" *PASTA:* " . . . will focus on the political headlines from last night . . . " **Change:** less structured formatting, but also weaker format adherence and less detail.
PASTA failure	Llama-3.1-8B / college_essays	$5/5 \rightarrow 2/3$	*Aligned:* "This dissertation aims to explore the evolution of traditional Chinese cultural values." *PASTA:* "the way we get these things, that's change" and "we can't be the same like them." **Change:** grammatical degradation and reduced coherence; topic remains recognizable.

Table 10: Illustrative aligned-vs-PASTA phrase-level contrasts from matched generations. The score column reports fluency/relevance judge scores. Rows 1–2 show high-quality PASTA outputs that preserve content while changing surface style; rows 3–4 show mild tradeoffs in polish, structure, or format adherence; row 5 shows a fluency/coherence failure. Aligned and PASTA outputs are independently generated continuations, not edits of one another.

G.3 Detector-specific extractors are most effective against their matching evaluators.

If scoring detector is known, using that same detector as the PASTA direction extractor always achieves the lowest or tied-lowest AI-detection rate among the extractors considered. In Table 11, the diagonal entries—where the extractor matches the evaluator—are underlined in all 12 model–evaluator columns. This pattern is strongest for Gemma-2-9B, where the matched extractor is strictly best for Pangram, FDG, and ImBD, and tied-best for Binoculars. The same trend also holds for Llama-3.1-8B and Qwen2.5-7B, where the matched extractor is consistently among the tied-best rows for every evaluator.

G.4 Contribution of different posttraining stages in AI-sounding generations

Table 13 shows that SFT is generally the stage where the PASTA-removable AI-sounding signature is strongest, but later stages can still contribute in specific domains. For the Llama/Tulu family, SFT yields large Pangram reductions across domains, including $+51.0$ for college essays, $+56.0$ for creative fiction, and $+59.1$ for scientific abstracts. The main exception is news articles, where PASTA remains effective even after SFT stages in both model families: Pangram reductions are positive for DPO and Instruct/RLVR in OLMo ($+3.0$, $+11.0$) and Tulu ($+12.0$, $+28.3$). OLMo also shows a later-stage effect in scientific abstracts, where DPO and Instruct exceed SFT under Pangram ($+22.0$ and $+14.0$ vs. $+9.4$). Overall, the per-domain breakdown suggests that SFT introduces the most stable detector-visible posttraining signature, while DPO and RLVR/Instruct add more domain-dependent contributions, especially for news and scientific abstracts.

G.5 Similarity of PASTA directions

Cross-model similarity of PASTA directions. Figure 16 compares PASTA directions across aligned models using a semantic logit-lens fingerprint: each model's direction is mapped through its own unembedding, projected into a shared MiniLM-L6-v2 semantic space, and compared by cosine similarity. The heatmap shows that PASTA

Table 11: Per-model AI-detection rate (%) on the $n = 125$ test set when different detectors are used as the PASTA extractor. Rows denote the detector used for PASTA direction selection; columns denote the detector used to score the resulting generations. **Bold** marks the canonical Pangram extractor row. Underline marks the lowest detection rate within each model and scoring-detector column; ties are all underlined.

Model	Scoring Detector → Direction Extractor ↓	Pangram	FDG	Binoculars	ImBD
Llama-3.1-8B	**Pangram**	**29.6**	**91.4**	**75.0**	**81.1**
	FDG	79.2	21.9	0.0	38.9
	Binoculars	79.2	21.9	0.0	38.9
	ImBD	79.2	21.9	0.0	38.9
Gemma-2-9B	**Pangram**	**1.6**	**96.2**	**72.0**	**60.0**
	FDG	26.5	16.5	0.0	4.3
	Binoculars	64.8	19.3	0.0	9.2
	ImBD	29.9	18.2	0.8	0.9
Qwen2.5-7B	**Pangram**	**13.6**	**92.9**	**71.2**	**85.6**
	FDG	98.4	89.7	41.6	99.3
	Binoculars	98.4	89.7	41.6	99.3
	ImBD	13.6	92.9	71.2	85.6

Table 12: AI-detection rate (%): in-domain vs. domain-agnostic PASTA, per (model, domain). (%); **bold** = the lower AI-detection value; cells where the two values are within 0.5% of each other are shown in *italics* on both sides (tie).

Model	Domain	Pangram	GPTZero	Originality	Fast-DGPT	Binoculars	ImBD
Llama-3.1-8B	college_essays	**16.0** / 20.0	**33.3** / 40.9	**84.0** / 91.8	**95.8** / 97.4	**76.0** / 88.0	**74.6** / 76.8
	creative_fiction	68.0 / **56.0**	*51.1 / 51.1*	**76.7** / 87.9	100.0 / **96.5**	*96.0 / 95.8*	99.8 / **98.7**
	news_articles	20.0 / **8.0**	**6.7** / 19.6	**65.7** / 84.0	95.3 / **77.1**	72.0 / **44.0**	91.6 / **60.6**
	opinion_pieces	*60.5 / 60.0*	**20.5** / 34.6	*96.0 / 95.8*	*98.2 / 98.4*	**88.0** / 92.0	96.7 / **93.1**
	scientific_abstracts	**0.0** / 4.0	38.5 / **26.5**	**37.2** / 58.3	**76.4** / 87.4	**20.0** / 56.0	**68.4** / 77.2
Qwen2.5-7B	college_essays	40.0 / **4.0**	**7.6** / 9.4	**41.5** / 88.2	**98.5** / 99.4	**72.0** / 92.0	**90.1** / 91.9
	creative_fiction	52.0 / **20.0**	**13.3** / 14.8	**80.4** / 95.5	99.0 / **98.2**	**80.0** / 88.0	97.7 / **94.4**
	news_articles	**12.0** / 24.0	**2.4** / 2.9	**76.2** / 95.3	94.6 / **91.9**	**28.0** / 48.0	71.8 / **61.7**
	opinion_pieces	44.0 / **8.0**	8.6 / **3.7**	**96.1** / 99.6	96.3 / **93.7**	**44.0** / 60.0	88.1 / **82.0**
	scientific_abstracts	*12.0 / 12.0*	11.1 / **9.0**	**28.5** / 61.4	**71.7** / 81.0	**20.0** / 68.0	**62.5** / 98.0

often recovers a shared behavioral direction across model families rather than a model-specific artifact. In particular, Llama-3.1-Instruct and the Tulu SFT/DPO/RLVR checkpoints form a highly coherent cluster, with pairwise similarities in the 0.91–0.98 range. The main exception is Qwen2.5-1.5B, whose direction is less aligned with the rest, while models at roughly $\geq$ 7B parameters show much stronger cross-family similarity. This suggests that PASTA identifies a common post-training behavioral fingerprint that becomes more stable at sufficient model scale.

Similarity among in-domain PASTA directions. Figure 17 measures cosine similarity among domain-specific PASTA directions extracted separately for each of the five domains. The heatmaps show that in-domain directions are never anti-aligned; all cosine similarities are positive and remain at least about 0.4, indicating that the directions share a common post-training component. At the same time, the directions are not identical: the maximum off-diagonal similarity remains below about 0.87, suggesting that each domain also contributes domain-specific structure. Opinion pieces appear especially central, with their in-domain direction relatively similar to directions from the other domains across the shown models. Together with the mixed in-domain-vs. domain-agnostic detection results in Figure 15, this suggests that domain-specific PASTA directions lie in a shared post-training subspace while retaining enough domain variation to produce detector- and domain-dependent wins.

Layer-specific ablation versus canonical PASTA. Table 14 compares PASTA against a layer-specific variant that ablates a different direction $v_{\text{cross}}[L]$ at each corresponding layer. For Llama-3.1-8B and Qwen2.5-7B, PASTA ablation is consistently better than layer-specific variant: for example, Pangram detection is much lower under canonical PASTA

Table 13: **Per-domain Δ in AI-detection rate (% absolute) across post-training stages.** Each cell reports three detector Δ values in the fixed order **Pangram / ImBD / Binoculars**, where Δ := (aligned AI-detection rate) − (PASTA-ablated AI-detection rate), computed at $n = 100$ generations per (family, stage, domain) slice. Positive Δ indicates that PASTA reduced the detector's flagging rate at that stage, while negative Δ indicates that PASTA increased it; **larger positive values therefore mean a stronger PASTA-removable AI-sounding signature.**

		Post-training stage Δ (Pangram / ImBD / Binoculars), %		
Family	Domain	SFT	DPO	Instruct / RLVR
Olmo-3-7B	College Essays	−6.1/0.0/ − 4.0	0.0/ + 2.0/ − 19.0	+1.0/ + 1.0/ − 20.0
	Creative Fiction	−2.0/ + 2.0/ + 3.0	+2.0/ + 1.0/ + 2.0	+3.0/ + 1.0/ − 3.0
	News Articles	−23.3/ − 8.8/ − 4.0	+3.0/0.0/ − 9.0	+11.0/ + 7.0/ − 15.0
	Opinion Pieces	−4.0/0.0/ + 8.0	0.0/ + 4.0/ − 1.0	0.0/ + 3.0/0.0
	Scientific Abstracts	+9.4/ + 1.0/ − 2.3	+22.0/ + 2.0/ − 7.0	+14.0/0.0/ − 23.0
Llama-3.1-8B & Tulu-3-8B	College Essays	+51.0/ + 2.0/0.0	0.0/0.0/ − 1.0	0.0/0.0/0.0
	Creative Fiction	+56.0/ + 4.0/ + 12.0	0.0/0.0/ + 4.0	+1.0/0.0/ − 2.0
	News Articles	+12.3/ + 19.0/ + 1.5	+12.0/ + 3.0/ − 8.0	+28.3/ + 2.0/ − 22.0
	Opinion Pieces	+36.0/ + 7.0/ + 19.0	0.0/0.0/0.0	0.0/0.0/ − 2.0
	Scientific Abstracts	+59.1/ + 17.5/ + 24.2	+4.0/0.0/ − 6.0	+1.0/0.0/ − 16.2

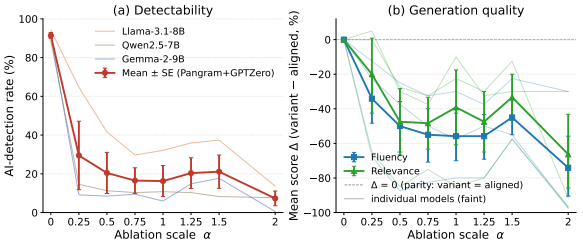


Figure 13: **Scaling PASTA exposes a detectability–quality frontier: low ablation strengths already sharply reduce AI-detection rates, whereas aggressive scaling further lowers detectability at the cost of fluency and prompt relevance.** Panel A reports AI-detection rate as a function of ablation scale α; error bars denote standard error across models, and faint lines show individual model trends. Panel B reports generation-quality differences relative to the aligned baseline for fluency and prompt relevance using pooled pairwise LLM-judge scores; the dashed line marks parity with aligned generations. **Lower values are better in Panel A**, while **higher values are better in Panel B**. Thus, the detector-minimizing α should not be interpreted as the quality-preserving operating point.

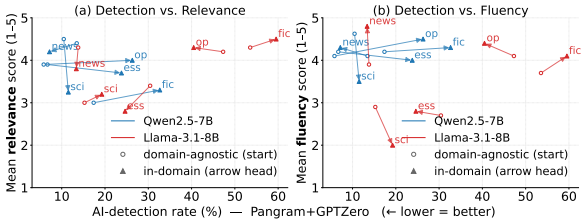


Figure 14: **Domain-specific PASTA does not provide a robust Pareto gain over domain-agnostic PASTA.** Each arrow points from the domain-agnostic generation (open circle) to the corresponding domain-specific generation (filled triangle); the Pareto-best direction is **upper-left**, indicating lower AI-detection and higher generation quality.The x-axis reports mean AI-detection rate across the two primary detectors, Pangram and GPTZero. Panel A compares detection against mean Claude Sonnet 4.6 **prompt-relevance** score, and Panel B compares detection against mean **fluency** score, both on a 1–5 scale. Arrows scatter rather than moving systematically upper-left: domain-agnostic PASTA has lower mean primary-detector detection in 6/10 model–domain cells, while domain-specific PASTA is lower in 3/10 with 1 tie, and neither quality metric differs significantly between the two variants.

than under per-layer ablation for both Llama (29.6% vs. 85.5%) and Qwen (13.6% vs. 92.8%), and the same pattern holds across GPTZero, Originality, FDG, Binoculars, and ImBD. Gemma-2-9B is the opposite numerically, with per-layer ablation lowering detection across detectors, such as Pangram 0.8% vs. 1.6% and ImBD 25.2% vs. 60.0%. Overall, Table 14 supports the canonical PASTA design: a single shared direction is sufficient and more robust for Llama and Qwen, while more aggressive layer-specific removal can reduce detectability by damaging generation quality.

Table 14: AI-detection rate (%) under per-layer direction ablation vs. canonical PASTA, across 5 detectors and 3 aligned models. Each cell shows "per-layer / canonical PASTA". Lower ⇒ less AI-detected; the **bold** number in each cell marks the lower (better) of the two and therefore the winning intervention for that (detector, model) pair.

Detector	Llama-3.1-8B	Gemma-2-9B	Qwen2.5-7B
Pangram	85.5 / **29.6**	**0.8** / 1.6	92.8 / **13.6**
GPTZero	45.6 / **34.5**	**8.6** / 10.3	57.2 / **7.9**
Originality	90.1 / **83.5**	**33.7** / 73.3	93.4 / **88.0**
FDG	94.1 / **91.4**	**72.5** / 96.2	98.5 / **92.9**
Binoculars	76.0 / **75.0**	**29.6** / 72.0	88.8 / **71.2**
ImBD	98.2 / **81.1**	**25.2** / 60.0	99.4 / **85.6**

Algorithm 1 Cross-Model Direction Extraction

Require: Aligned model $\mathcal{M}_{\text{a}}$; Base model $\mathcal{M}_{\text{b}}$; aligned-format prompts $\mathcal{P}_{\text{a}}$; base-format prompts $\mathcal{P}_{\text{b}}$ ($|\mathcal{P}_{\text{a}}| = |\mathcal{P}_{\text{b}}| = N$); generation hyperparameters T; layer count L_{tot}; residual width d.

Ensure: Unit-norm directions $\{v_{\text{cross}}[\ell]\}_{\ell=0}^{L_{\text{tot}}-1}$.

for $(\mathcal{M}, \mathcal{P}) \in \{(\mathcal{M}_{\text{a}}, \mathcal{P}_{\text{a}}), (\mathcal{M}_{\text{b}}, \mathcal{P}_{\text{b}})\}$ **do**
  $s[\ell] \leftarrow \mathbf{0} \in \mathbb{R}^d$ **(fp64)** for $\ell = 0, \ldots, L_{\text{tot}} - 1$; $n \leftarrow 0$
  for each prompt $p \in \mathcal{P}$ **do**
    $g_p \leftarrow \textsc{Generate}(\mathcal{M}, p, T)$ ▷ natural continuation
    $\text{ids} \leftarrow \text{concat}(\textsc{Tok}(p), \textsc{Tok}(g_p))$
    $H \leftarrow \textsc{Forward}(\mathcal{M}, \text{ids}; \text{hidden_states})$ ▷ $H[\ell] \in \mathbb{R}^{|\text{ids}| \times d}$
    $\text{st} \leftarrow |\textsc{Tok}(p)| - 1$; $\text{en} \leftarrow \text{st} + |\textsc{Tok}(g_p)|$
    for $\ell = 0, \ldots, L_{\text{tot}} - 1$ **do**
      $s[\ell] \leftarrow s[\ell] + \sum_{i=\text{st}}^{\text{en}-1} H[\ell][i]$
    end for
    $n \leftarrow n + |\textsc{Tok}(g_p)|$
  end for
  $\mu_{\mathcal{M}}[\ell] \leftarrow s[\ell]/n$ for $\ell = 0, \ldots, L_{\text{tot}} - 1$ ▷ per-model mean
end for
for $\ell = 0, \ldots, L_{\text{tot}} - 1$ **do**
  $\Delta[\ell] \leftarrow \mu_{\mathcal{M}_{\text{a}}}[\ell] - \mu_{\mathcal{M}_{\text{b}}}[\ell]$
  $v_{\text{cross}}[\ell] \leftarrow \Delta[\ell]/\|\Delta[\ell]\|_2$
end for
return $\{v_{\text{cross}}[\ell]\}_{\ell=0}^{L_{\text{tot}}-1}$

Algorithm 2 Inference-Time Directional Ablation

Require: Aligned model $\mathcal{M}_{\text{a}}$ with L_{tot} transformer blocks; unit direction $v \in \mathbb{R}^d$ with $\|v\|_2 = 1$; ablation strength $\alpha \geq 0$ with default $\alpha = 1$; prompt p; generation hyperparameters T.

Ensure: Detection-ablated generation g.

for $\ell = 0, \ldots, L_{\text{tot}} - 1$ **do** ▷ register $3L_{\text{tot}}$ hooks
  pre-hook on block ℓ: $x \leftarrow x - \alpha(x \cdot v)\, v$
  post-hook on attention output of block ℓ: $x \leftarrow x - \alpha(x \cdot v)\, v$
  post-hook on MLP output of block ℓ: $x \leftarrow x - \alpha(x \cdot v)\, v$
end for
$g \leftarrow \textsc{Generate}(\mathcal{M}_{\text{a}}, p, T)$ ▷ hooks active throughout decoding
Remove all hooks from $\mathcal{M}_{\text{a}}$
return g

Algorithm 3 Layer Sweep for Identifying ℓ^*

Require: Aligned model $\mathcal{M}_{\text{a}}$; directions $\{v_{\text{cross}}[\ell]\}_{\ell=0}^{L_{\text{tot}}-1}$ from Alg. 1; evaluation prompts $\mathcal{P}_{\text{eval}}$; generation hyperparameters T; AI-text detector $\mathcal{D}$ (e.g., Pangram).

Ensure: Causally most effective layer ℓ^* and sweep $\{d[\ell]\}_{\ell=0}^{L_{\text{tot}}-1}$.

for $\ell = 0, \ldots, L_{\text{tot}} - 1$ **do**
  $\mathcal{G} \leftarrow \emptyset$
  for each prompt $p \in \mathcal{P}_{\text{eval}}$ **do**
    $g \leftarrow \textsc{AblatedGenerate}(\mathcal{M}_{\text{a}}, v_{\text{cross}}[\ell], p, T)$ ▷ Alg. 2
    $\mathcal{G} \leftarrow \mathcal{G} \cup \{g\}$
  end for
  $d[\ell] \leftarrow \mathcal{D}(\mathcal{G})$ ▷ mean fraction flagged AI
end for
$\ell^* \leftarrow \arg\min_{\ell \in \{0, \ldots, L_{\text{tot}}-1\}} d[\ell]$
return ℓ^*, $\{d[\ell]\}_{\ell=0}^{L_{\text{tot}}-1}$

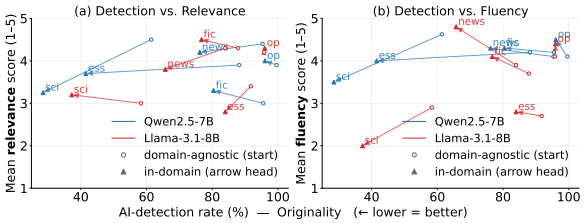
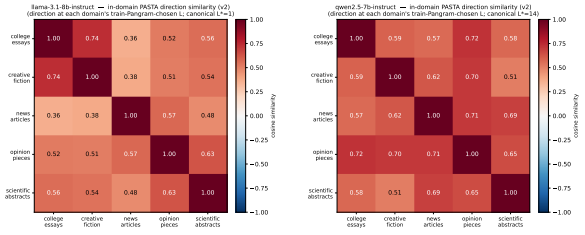


Figure 15: **Originality is a detector-specific exception: in-domain PASTA often lowers detection, but does not yield a consistent quality gain.** Each arrow corresponds to one model–domain pair and points from the domain-agnostic generation (open circle) to the corresponding in-domain generation (filled triangle); the Pareto-best direction is upper-left. The x-axis shows AI-detection rate under Originality, while the y-axis shows mean Claude Sonnet 4.6 relevance score in Panel A and fluency score in Panel B. Most arrows move left, consistent with Appendix Table 12, where in-domain PASTA lowers Originality detection in 9/10 model–domain cells, but arrows do not consistently move upward, indicating that the Originality-specific detection gain is not accompanied by a reliable quality improvement.



(a) Llama-3.1-8B (b) Qwen2.5-7B

Figure 17: Cosine similarity between in-domain PASTA directions $v^{D}_{\text{cross}}[L^{\star}_{D}]$ across the 5 domains, per model. Each heatmap is a 5×5 symmetric matrix over the in-domain directions extracted at the PASTA-chosen layer $L^{\star}_{D}$ for each domain.

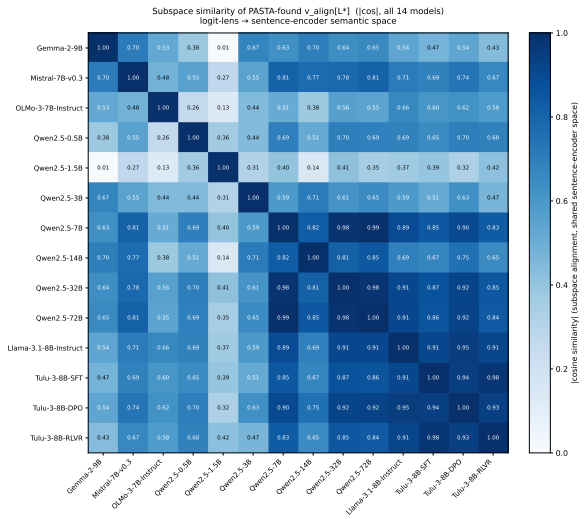


Figure 16: Pairwise cosine similarity between the behavior of PASTA ablation directions across 14 aligned models, computed via the semantic logit-lens fingerprinting procedure. Cell (i, j) reports the absolute cosine similarity between the semantic-space behaviors of models i and j.